OPTION MARKET MAKING

Trading and Risk Analysis for the Financial and Commodity Option Markets

ALLEN JAN BAIRD

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Preface

Since the introduction of stock option trading in 1974 and futures option trading in 1982, the total volume of option trading on all exchanges and over-the-counter markets in the United States is now billions of dollars a day. Option trading is one of the largest and most rapidly growing sectors of the financial industry. After a lag of about a decade, the option markets of Europe and Asia are following the growth path of the United States, promising that world option markets will become one of the most important sectors of the global financial industry in the 1990s and probably into the next century.

This book enumerates and evaluates the basic risks, strategies, and tactics of option market making as a profitable business. Its goal is to be both a theoretical and a practical reference for option traders, dealers, and market makers in financial and commodity option markets, whether they are trading on the exchange floor or in the growing national and international over-the-counter option markets.

Option market makers who will find this book of interest include dealers at banks and securities firms and trading members of option exchanges and over-the-counter markets. Interest rate managers in insurance and industrial corporations, investment officers of pensions, trusts, or funds, and option speculators should also find that knowledge of market making will improve their trading results or objectives.

The success of financial option markets requires the participation of many different traders: commercial hedgers, investment funds, speculators, and dealers or market makers. The functions of the first three are well known, but the role of the option dealer or market maker has not been widely researched, even though on a daily basis market maker trading volume is about half of all option trading. There are many good books about option trading but few, if any, exclusively address option market making.[1]

[1]Although there have been no books previously published about option market making, recently several introductory articles or chapters in books dealing briefly with this topic have appeared (CBOE, 1990; Colburn, 1990; Silber, 1988).

With a lack of good materials about this trading activity, option market makers learn their trading skills either from their employers or partners if they are lucky, or by trial and error, usually losing money in the process.

In the hope of making "paying one's dues" less costly for new traders, this book offers for the first time a comprehensive reference exclusively devoted to option market making in the commodity, bond, stock and currency option markets. Particular attention is given to futures options markets.

The function of market makers is to offer the service of *immediacy,* by always being willing to buy or sell for some quantity any option at some specified price. Thus, they provide liquidity and stability to the option markets. In return for liquidity and stability, market makers try to earn as income the difference in the price spread between bid and offer at which they will trade. In this respect, the profit function of market making is like any business in which merchandise must sell at a higher price than the cost of acquisition; but there are specific business risks to market making that must be thoroughly understood. Successful market makers may earn a high income as a percentage of capital invested, if a large turnover with low or containable risk can be assured. Because of the close relationship between risk and income, option risks and how they are managed are at the heart of profitability and success in market making.

The author is a professional option market maker on a futures option exchange, and this book largely stems from this professional experience. Professional floor traders generally content themselves with trading rather than writing. Indeed, publicizing trade secrets may be viewed as a self-defeating endeavor. There is probably some truth in this observation. Nevertheless, publicizing trade secrets will probably not change the fundamental profitability of option market making nearly so much as one might imagine. In any case, trade secrets rarely remain permanently secret.

STRUCTURE OF THIS BOOK

Chapters 1 and **2** provide an overview of the economics of market making and review option basics and terminology. Although the reader should have previously read at least one elementary

book about options, some review of basic option definitions and theory is given here. This includes an introduction to fair value models, volatility, and some important differences between commodity, bond, currency and stock options markets.

Chapter 3 formally takes up the question of option risks and how they are to be measured. Trading options involves far more risks than other investments, and more complex risks at that. All option traders must be alert to the variety of these risks in daily trading. These risks are defined and summarized.

Chapter 4 explores different possible option strategies and the various risks in single-month option trades. It is always important to know the risk profile of any option position so that one can distinguish between those that risk financial catastrophe and those that promise safe and secure profits. Ignorance of risk may lead to financial ruin or sharp setbacks in trading as well as market making.

Chapter 5 discusses option market making from the perspective of synthetic option trading. This form of market making is virtually risk-free and represents the core of option arbitrage trading. Mastering synthetic price relations is essential for good scalping profits.

Chapter 6 systematically addresses the question of calendar spread risk and strategy. Understanding time risks is essential if one is to avoid the possibility of a catastrophic loss of market maker or public trading capital. Core market making trading strategy comes out of understanding how to limit time spread risk but still use time markets efficiently.

Chapter 7 discusses the overall strategic considerations in nonsynthetic market making from several perspectives. Topics include delta neutrality, neutral time spreading, broker order flows and net supply and demand, trading fences, and managing option cycle expirations. Detailed attention is given to the butterfly/ratio time spread strategy under different market conditions.

Chapter 8 takes up certain practical matters and tactics in making option markets in an active trading environment. Making a spread market, risk adjustment, gamma trading, implied volatility, skew, tracking, financial results, and avoiding mistakes are some of the topics.

Chapter 9 includes some personal observations about floor trading as a business and social experience.

Option market makers almost always use some option software to keep track of and analyze their option positions while trading. The reader does not need such software to understand this book, but active traders will almost certainly use option software. An appendix provides recent software information on option market making.

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OPTION MARKET MAKING

1

Economics of Option Market Making

This study describes and evaluates the risks and opportunities in option trading from the standpoint of the professional option market maker. A market maker is a trader of options or any other liquid financial asset who is willing and ready at any time to buy at a bid price or sell at an offer price whatever is being traded. Market makers may also be known as dealers, especially in the securities, bond and currency over-the-counter markets, and sometimes as scalpers, when both the buy and sell sides of the trade are expected to be completed within a very short time. In stock and stock index option markets, market makers are institutionalized in the specialist system, but in futures options markets, market makers are independent traders in open competition.

A market maker provides a ready and liquid market for publicly brokered orders and, thus, the service of immediacy. The flow of buy-and-sell orders is rarely evenly matched at any given time, so without market makers prices would become more volatile and erratic as commercial and public buy-and-sell orders would become imbalanced (that is, another trader would not be willing to assume the opposite risk at that moment). By stepping into the market to meet the otherwise unfilled orders, market makers help bring prices back to a fair-value level that more accurately reflects the true supply/demand equilibrium. Market makers do not set prices (these are determined by more fundamental factors affecting supply and demand), but they do reduce price volatility, assuring the public of better and more accurate prices while trading in open markets. Market-making dealers are found on all financial, futures, and

over-the-counter option markets and are probably essential to the functioning of any open and fair market in which liquidity is important.

The above market situation may easily be reflected with the elementary supply/demand curve shown in Figure 1.1. Consider a situation in which a long-term equilibrium price is established at P* for some tradable financial asset. If a temporary imbalance of demand pushes the price to P1 or higher, market makers may be expected to enter the market and offer for sale a sufficient quantity to bring prices back to equilibrium. Likewise, if prices were to fall temporarily to P2 because of an imbalance between public buyers and sellers, market makers may be expected to enter the market and buy sufficient quantities to bring prices up to long-term equilibrium. At some level prices P1 and P2 around the equilibrium price will become the *asked* and *bid* prices at which a market maker is willing to trade. A more complete supply/demand analysis of market-maker services for stocks and a review of previous work may be found in Stoll (1987).

For every buyer or seller of options (or futures), there is an opposing seller or buyer. A long option holder is always paired with a short option seller (or writer). Thus, on a single option or futures contract, what one trader makes in profit the other must lose. Considering the option and futures market only, option and futures trading is a zero-sum game. Futures and futures option exchange trading neither adds nor subtracts from total wealth directly but shifts it from one group to another.

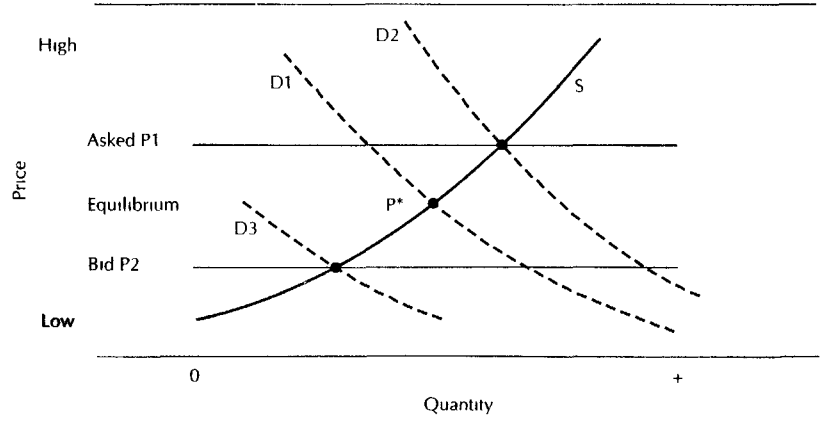

Figure 1.1 Dealer price.

Economic activity that leaves total wealth unaltered or merely shifts existing wealth is not necessarily useless or harmful, however. Insurance, for example, does not directly create wealth; but it provides a socially useful service of diversifying and shifting the risk of large and unexpected losses from one group to another. It is this risk-shifting and insurance function that futures and options markets provide to industry.

As an example, commercial firms, free of commodity price change risk with hedged futures, can concentrate on productive economic activity more rationally, and thereby increase total wealth. Although futures and options trading do not directly add to or subtract from total wealth, they provide a price insurance service to primary product producers and industrial consumers, thus increasing productivity and thereby total wealth.

Within the risk perspective of futures markets, options on futures represent the extension of risk shifting opportunity for futures speculators. With options, a speculator or hedger may insure or hedge his or her own futures holdings, either to lock in profits or to hedge against adverse futures price movements. Options are forms of risk shifting for the futures market itself. They are one way for commercial hedgers or speculators to insure their own futures risk.

For futures markets to perform price discovery and risk shifting functions successfully, futures market participants must include other market traders in addition to commercial hedgers. These traders are scalpers, market makers, and speculators. Scalpers and market makers assume that portion of risk that commercial hedgers or speculators wish to shift temporarily, but eventually wish to neutralize. Speculators ultimately assume the risk that commercial hedgers and market makers do not want. In effect, market makers and speculators distribute the unwanted investment risk of commercial hedgers and perform an insurance function to industry.

For providing a service of immediacy and liquidity, of course, a market maker expects to earn some profit from the difference between the temporary disequilibrium price and the long-term equilibrium price. By buying at the bid price and selling above long-term equilibrium at the asked price, the market maker earns this difference as profit for providing liquidity service. Buying low and selling higher is the cost framework of any business, and in exchange-floor terminology, this is known as "getting the edge."

Getting the edge is very important because, all else being equal, the market maker earns this price spread as income and profit. When a trader earns both sides of the edge on one contract, the trade is liquidated. Studies of securities dealers and futures scalpers suggest that the market makers are unlikely to earn the entire bid/offer spread as net profit on every trade, and this fact is true for option market makers as well (Working, 1977; and Silber, 1984 and 1988). There are certainly costs and risks to market making that reduce gross profits. As exchange members or over-the-counter dealers, market makers pay very low trading or brokerage fees; but they also must pay other overhead charges against memberships, office space, clerical and data services, and other business costs. Trading is not entirely cost-free even for market makers, and gross profits of the price spread will thus be reduced. The minimum price spread at which a market maker may expect to make a profit is known as the *reservation price.*

The difference between getting the edge and the reservation price represents the reduced gross profit, but even this is not necessarily net profit. Market making is not risk-free by any means. In earning a liquidity function income, market makers, dealers, and scalpers are exposed to the risk of windfall loss. The obvious risk to any scalper is that he or she will not be able to complete both sides of a bid/offer spread fast enough to earn more than the reservation price. A scalper may buy at the bid price but be unable to sell at the offer price. If the scalper sells the contract to another trader or broker at the same bid price at which the scalper bought the contract, the scalper would break even ("scratch the trade").

If the scalper was able to buy at his or her bid price, but unable to sell at either the original offer or the reservation price, then the scalper incurs an outright loss. Of course, in the interval between buying at the bid price and selling at the offer price, prices could rise, giving the scalper a windfall profit greater than the difference in the original bid/offer spread. This element of investment risk negates any guarantee that market making will be profitable as a business.

To maximize profits, market makers ideally would like to make the bid/asked dealer spread as wide as possible, but the dealer spread will be influenced by the demand for liquidity services and the supply of service or competition among market makers. The more actively traded the market and the greater the number of market makers, the narrower the dealer spread. In highly sat-

urated markets, the presence of too many market makers will quickly narrow the dealer spread to the reservation price itself, in which case very little profit is earned by market making. Of course, this competitiveness among market makers gives broker orders the best possible price of execution. Indeed, to gain the edge on one trade, a market maker will often give up the edge in another trade, resulting in broker orders getting filled at better than equilibrium price to the benefit of the public trader.

Although the dollar profit on any individual trade may be small and usually is accompanied by numerous scratch or small-loss trades, the cumulative earnings from market making may become large in high- and active-volume markets when market making is done well. Of course, scalpers must always face the possibility of windfall losses as well as profits. Market making is not risk-free, and much market-making trading strategy is directed to investment risk control.

Differences in the degree of risk between dealing in securities, bonds, currencies, or futures markets and in option market making tend to affect dealer prices somewhat differently in option markets than in other asset markets. One difference between options and assets scalping, for example, is that market instrument liquidity in options trading is lower than in trading of other financial assets.

Stocks, currencies, and bonds are single-asset instruments, whereas there is no one option that matches the underlying risks of those assets. The multitude of strikes and calendar series (as defined in Chapter 2) in which options trade may easily push the distinct options traded into the hundreds for each underlying instrument, many of which may not trade on a regular basis. While an underlying-asset market maker is always a scalper on a short-term basis, this observation is not necessarily true of options market makers.

Although some near-term, near-strike options may trade in active markets almost as frequently as futures contracts, many options by month or strike may trade only infrequently. Thus, the market maker may end up owning or shorting contracts that he or she must be prepared to hold for long periods of time (even to expiration) unless he or she wishes to liquidate to other market makers, thus giving up the edge. Almost invariably, therefore, options scalpers must assume carryover positions on a regular basis, something a futures scalper does not do. It is a rare option market maker who is able to go home at the end of the day completely

flat without overnight positions. Option market makers assume a large carryover risk that other financial dealers may not, and this longer-term assumption of risk may affect dealer price.

Option risk is more complex than many other investment risks. In assuming long-term risk exposure in the option market, market makers must be careful to understand exactly what these risks are. Options are highly leveraged instruments that do not always move in value dollar for dollar exactly corresponding to a futures or cash asset price change. The values of options characteristically shrink or explode at rapidly changing and sometimes greater rates than the underlying instrument. Options tend to decline in value over time (all else being equal), but option prices can also increase dramatically even without any time passing or change in futures price at all.

In being forced to hold long-term carryover positions in options, and thereby in being exposed to some long-term complex investment risks, market makers may lose in carryover positions what they make in liquidity function profits. In the worst case, a market maker can lose his or her entire capital. The 1980s witnessed perhaps half a dozen well-publicized multimillion dollar bankruptcies of large option traders, including two commodity exchange clearing houses—Volume Investors and Fossett (*The Wall Street Journal,* Oct. 13, 1987, and Michael Siconolfi, Nov. 12, 1990; *The New York Times,* Oct. 18, 1989). Although these bankruptcies were not necessarily those of option market makers, they clearly demonstrate the potential risks of option trading by the unwary and uninformed.

In attempting to earn a $10 liquidity function profit, option scalpers risk watching their long inventory go to zero and their short inventory rise to stratospheric prices in short order. This possibility represents catastrophic loss and is the most serious risk to any option trader. Floor traders who go bankrupt, or "blow out," trading options are a grim reminder that there are no easy profits in option trading.

An option market maker must always be prepared to "marry an option" (that is, to hold a long-term option position without liquidation or getting out), even as he or she strives to make a liquidity function profit on the transaction without taking undue risk. Since the chief risk is the possibility of financial catastrophe in the option carryover position, much of the skill in option market

making comes from being able to manage these additional risks well in a large carryover position until expiration.

The perspective of this study is pure market making, where income comes solely from market making and not from speculative gain (or loss). Market-making, risk-neutral trading goals are to make a profit, not to be proved right about market direction.

To develop a prudent option market-making strategy, one must understand catastrophic option risk and establish practical measures to avoid this exposure; otherwise, one risks bankruptcy over the long run. Fortunately, not all option trading positions are catastrophically risk exposed, and some are limited-risk strategies. Limited-risk strategies for option market makers are not only safer than other option strategies, but also they are often safer than trading in the underlying asset itself. Knowledge of risk is the key to market-making option strategy.

Limited-risk strategy does not mean limited profits. Many low-risk strategies will also earn among the highest possible rates of return of any option strategy. Although market making as an economic activity earns an income for providing market liquidity, it is also possible to make option markets that sometimes earn large profits from time to time on the carryover position in windfall situations.

2

Options

WHAT IS AN OPTION?

> A call (put) option is the contract right to buy (sell) a specified amount of some real or financial asset at a fixed price on or before a given date.

If the option purchaser acts upon this right to buy, he or she is *exercising* this right; and the fixed price of the transaction is known as the *strike* price. The seller of the option, known as the *writer,* must be prepared to sell the specified asset when the option purchaser exercises these rights. When the option buyer exercises, the seller is *assigned.* The *maturity* of the contract is known as the *expiration* date, and exchange option trading takes place in any one of a number of set contract months, or *cycles.* An *American* option allows the holder to exercise the right any time before the expiration, and a *European* option restricts the right only to expiration and not before.

A call is *in the money* at expiration when its asset price is above the exercise price, and a put is in the money when the asset price is below the exercise price. A call is *out of the money* at expiration when its asset price is below the exercise price, and a put is out of the money when the asset price is above the exercise price. Options in the money have real value, and those out of the money have no remaining value at expiration. An option is *at the money* when the asset price expires or trades right at the strike.

The underlying *asset* may be either the cash market (spot) or futures contracts on the underlying spot instrument (stocks, bonds, currency, or spot commodities). The spot option market holds a right to exercise over the cash asset itself. The futures

option is on the futures contract only, which is then a right to delivery of the spot instrument. There may be both a cash and a futures option market trading in the same asset market, in which case cash/futures option arbitration will become important, whether in the stock, bond, and currency futures or over-the-counter markets.

The gross profit/loss (P/L), or expiration, payoff of a $100 strike call and put is illustrated graphically in Figures 2.1 and 2.2, where the purchase or trade price of both the put and call is $1.70. The 100 call allows the long holder to exercise the right to buy the underlying asset at $100 on (or sometimes before) expiration, and the short call writer must fulfill this demand. The 100 put allows the long holder to sell the asset at $100 to the short put writer on or before expiration.

Profit/loss is based on the expiration value of the option minus its trade price. If the asset settles at expiration at $102, for example, the long 100 call will expire in-the-money valued at $2.00 for a profit of 30 cents ($2.00 – 1.70) and a 30-cent loss for the short call. At an asset price of $110 at expiration, the call will be worth $10.00, for a net profit of $8.30 for the long call and a loss of $8.30 for the short call. The breakeven point for both the long and short call is an asset price of $101.70 at expiration, based on the trade

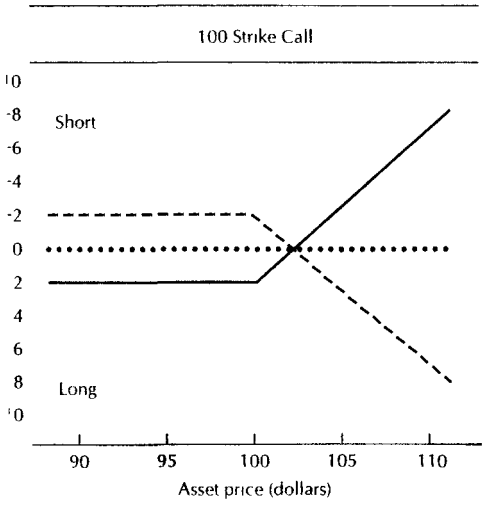

Figure 2.1 Payoff for long/short 100 strike options at expirations.

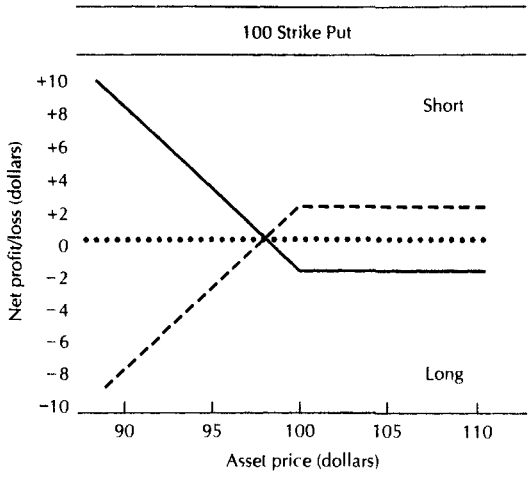

Figure 2.2 *Payoff for long/short 100 strike options at expirations.*

price of the option at $1.70. Below this asset value the long call will show a maximum loss of $1.70, and the short call will show a maximum profit of $1.70.

The breakeven point for the long or short 100 put will be an asset price of $98.30 at expiration, above which the long put will have a maximum loss of $1.70 and the short call a maximum profit of $1.70. Below the breakeven point, the long put will show increasing profit and the short call increasing losses.

While determination of the value of an option at expiration is relatively straightforward, its value before expiration is unknown. How much should an option be worth, or what is its fair value, before expiration? Clearly, it must equal at least the immediate exercise in- or out-of-the-money value or else an arbitrage profit generally becomes available, forcing price and value back in line. But how much more should be added above the in-the-money value to determine the fair value?

An options price depends on both the *intrinsic* and the *extrinsic* value of the option. The intrinsic value of an option is simply the value of the option if it were exercised immediately for cash value. The intrinsic value is always known and is a simple function of the relationship between asset price and the option strike. If the asset price is below the strike price of a call, the call will have no

intrinsic value; and if the asset price is above the strike of the call, then the intrinsic value is the positive difference between the asset price and the strike. Puts have intrinsic value if the asset price is below the strike, but no intrinsic value if the price is above the strike. Options that have positive intrinsic value are in the money, and those without intrinsic value are out of the money.

For example, if a May future is at $110, then the 100 May call will be in the money and will be worth exactly $10 in intrinsic value, because a trader can exercise his or her long call rights to buy futures at $100 and immediately sell them for $110 on the open market. If futures are selling below $100, however, the 100 call will have no intrinsic value and will expire with no worth if this relationship holds until expiration. At expiration an option will be worth exactly its intrinsic value.

Whether a position has a net profit at expiration will, of course, depend on what the trader paid for the call. Only if the trader bought the call for less than $10 will there be a net profit if the futures price is at or above 110 at expiration. If the futures price closes below 100, however, the trader will lose money no matter what he or she paid for the 100 call.

In theory, the intrinsic value of an option will always set the minimum price that an option must have, since if it fell below that value, an arbitrage profit for market makers would quickly close the gap. Although option prices are limited on the downside by their intrinsic value, in practice, they usually trade above this value as long as there is time remaining to expiration.

Remaining time to expiration will add some additional value to an option's intrinsic value, which is known as the extrinsic value or time premium. Generally, the longer the time remaining, the higher the time premium. This valuation makes intuitive sense since a longer time remains for the option to become intrinsic-value profitable and, therefore, the option must be more valuable than at expiration.

Although the intrinsic value of an option is a direct function of futures or asset prices, it is the addition of extrinsic value or time premium that sets the total value of an option. Option valuation is essentially time premium pricing. How does the market set the price of the extrinsic value of options? How much should the time premium of an option be worth? Answers to these questions will be developed in the next section.

OPTION FAIR VALUE

An option may be considered fairly valued when it is priced so that trading at that price will produce neither gain nor loss over the long run, whether trading from the long or the short side. It is fairly valued because each side of the trade has equal economic advantage. When options are fairly valued in the market, they are best able to shift risk away from the underlying asset market. Acting essentially as an insurance factor to the underlying asset market, market makers in options must evaluate the risks of price changes carefully in order to charge an appropriate price for the insurance. How should options be fairly priced?

Modern fair-value option models, based on statistical models of probability, attempt to provide the answer to this question. The worth of an option is related to how likely it is to earn a profit or loss. To know fair value, however, requires a trader to know the probability of gain and loss for any specific option and, therefore, the probable future underlying asset price as well.

Before modern fair-value options models were developed, no theoretical model of option pricing could successfully generate hypothetical fair values of option time premiums. The price risk exposure of an option contract was evaluated subjectively, with only past empirical price and time relations somewhat imperfectly known. One can still value options in a relative way without a fair-value model if one knows the fixed arbitrageable synthetic price relations between puts, calls, and futures or asset prices (see Chapter 5). But synthetic option price relations do not establish a model of fixed absolute levels, that is, fair value itself.

Modern stock option pricing theory dates from the early statistical models of option premiums, especially the work of Fischer Black and Myron Scholes (1973), which was modified by Robert Merton (1973) into the BSM model. This model was adapted to commodity futures options by Fischer Black. The BSM model quickly came to be the most widely recognized and used stock and futures option value model; and it provides the statistical basis of this study, although alternative models will also be discussed for bond markets.

To understand how options are fairly valued statistically, let us review the probability of any kind of payoff-game outcome. As an example, consider a game in which, after a trial of only one period,

there are only two possible outcomes, say heads or tails, with one outcome counting as a win and the other as a loss. Suppose a win results in a reward of \$5, and a loss results in no reward, or \$0. If the probability of either a win or a loss on one trial is 50 percent, what bet will neither gain to nor lose money if this game is played again and again? The answer to what a fair-valued bet should be is the probability of outcomes times the payoff, summed, or $.50 \times \$0 + .50 \times \$5 = \$2.50$. Out of every two trials, there will be a total gain of just \$5 on average (one win and one loss, or \$5 + \$0). Since it takes two trials on average to win \$5, each bet must be \$2.50 in order neither to win nor to lose money over a two-trial run over many plays.

The general formula for the fair value of a bet over a one-period trial, then, is the probability payoff schedule:

$$V = \sum(P) \cdot (\$)$$

where V = value of fair-value bet

P = probability of outcome

\$ = outcome payoff

Now consider a game in which there is only a one-period trial and for which the outcome values are either 105 or 95. Assume that an outcome of 105 will bring a profit of \$5 and an outcome of 95, \$0. Assume also that the probability of 105 or 95 occurring is exactly 50 percent on each trial. What is the value of a bet that will produce neither a profit nor a loss over the long run?

It may be seen that the answer again is exactly \$2.50. Both games just described represent a one-event, binomial distribution with identical outcome payoffs and event probabilities. Calling a win heads, or 105, and a loss tails, or 95, does not change the essential similarity of the two games in any way. In each case, the bet that will produce neither a win nor a loss is calculated as the sum of the individual outcome probabilities times the outcome payoff, that is, $.50 \times \$0 + .50 \times \$5 = \$2.50$.

Observe that the bet in the second game is equivalent to an investment or purchase of a hypothetical 100 call that can take only an exercise value of either 95 or 105 over a single period. By using elementary probability theory, one may calculate the fair value of a call option under these restricted specifications.

Now consider the possibility that outcomes are distributed within a range of 95 to 105 in one-point increments (that is, 95, 96,..., 105) and that the gain on each outcome is \$0 at or below 100, and the numerical outcome less \$100 when above 100; for example, 95 = \$0, 96 = \$0,..., 103 = \$3, 104 = \$4, 105 = \$5. This result is similar to the payoff schedule of a hypothetical 100 call taken over a one-trial event.

If the probabilities of the specific outcomes are known, then, as in the previous examples, it is possible to calculate that fair-value investment or bet that will produce neither profit nor loss in the long run. For sake of illustration, assume that the outcomes from 95 to 105 have the following probabilities of occurring over a one period trial:

Possible Outcomes	*Probability of Occurring (percent)*
95	2
96	4
97	8
98	12
99	14
100	20
101	14
102	12
103	8
104	4
105	2

These probabilities are graphed in Figure 2.3, where the mean strike is 100.

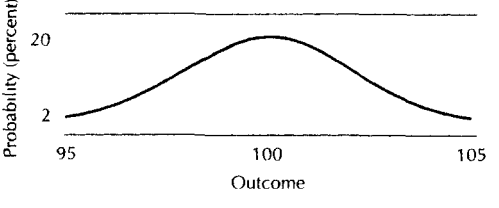

Figure 2.3 Outcome probability.

Since the specific payoffs as well as the probability for each outcome are known, the fair value of a bet may be determined exactly as in the previous games. That is, one multiplies the probability of an outcome by its expected payoff and sums the results to find the exact fair-value bet or investment. Thus,

Outcome 95	$0 × .02 =	$0
Outcome 96	$0 × 04 =	0
Outcome 97	$0 × .08 =	0
Outcome 98	$0 × .12 =	0
Outcome 99	$0 × .14 =	0
Outcome 100	$0 × .20 =	0
Outcome 101	$1 × .14 =	0.14
Outcome 102	$2 × .12 =	0.24
Outcome 103	$3 × .08 =	0.24
Outcome 104	$4 × .04 =	0.16
Outcome 105	$5 × .02 =	0.10
	Sum =	$0.88

Summing the expected payoffs gives a result of 88 cents, which represents the fair-value bet in this example, that is, that investment that neither wins nor loses repeatedly on one-period trials over the long run.

In the above example, one readily sees that the payoff schedule of each outcome (before the probability is taken into account) is just the payoff schedule of a hypothetical 100 call at expiration, or its intrinsic value from 95 to 105. Therefore, the only unknown in calculating the fair-value of such an option is the specific probability of each outcome. In other words, once the probability of outcome for an option's intrinsic value over some trial period is known, that option's fair-value price can be determined exactly.

In the preceding examples, we have assumed that the probability of outcomes is known exactly. But in the real world, the exact probability of occurrence for any specific intrinsic value of an option at expiration is not known directly. How, then, is it possible

to derive such outcome probabilities for option analysis? How can we know what the probabilities are for the payoff of a hypothetical 100 call from 95 to 105, or from 50 to 150? The answers to these questions will be discussed in the next section.

OPTION PRICING MODELS

The key element of any option fair-value model is the probability assumptions about changes in underlying asset prices. If the probability distributions of asset price changes are known or can be successfully estimated, then these may be used to derive the probability densities of the expected payoff schedule of options at expiration, from which fair value may be derived.

More generally, if it is theoretically known that an asset price has an x percent chance of increasing by y points or more over the next z days, then an option price will be related to the outcome of this price change probability. A fair-value option price only reflects the intrinsic value at expiration, which is linked to the probability of underlying asset price change. If the distribution of futures prices is accurately estimated, so too will be the fair value of the option.

The probability of asset price change is usually referred to as *volatility* by option traders, and it is usually unknown. As a consequence, volatility must be estimated. Doing so gives rise to a number of different option pricing models.

Two other statistical assumptions or unknowns must also be incorporated in order to derive fair value for any model. (1) The risk-free interest rate must be known or estimated over the life of the option; and, (2) if the option is on a yield bearing asset, the dividend or yield must be known or estimated over the life of the option. For bond options in particular, additional estimates of the term structure of interest rates may also be necessary. This chapter will review the Black-Scholes-Merton (BSM) option pricing model, but we shall also discuss some recent work in lattice-based and advanced bond option models.

The work of Black and Scholes in 1973, followed quickly by that of Merton the same year, proposed to link the probability of stock price changes to stock options using a log-normal distribution as the probability estimator. The BSM model was based on earlier

statistical work that had shown that stock price changes could be modeled on a normal or Gaussian curve (Cootner, 1964). That is, futures (and stock) price changes resemble a random sample drawn from a universe that can be described by a log-normally distributed curve.

The finding that stock and asset price changes resemble known probability distributions makes it possible to develop theoretical models of price changes. The earliest fair value model of Black and Scholes used the normal-curve model of futures prices to derive option prices. The BSM model deduces option prices on the basis of statistical theory and thus supercedes subjective or graphical option price valuation. For example, the Fischer Black (1976) formula for European futures option fair value is:

$$C = e^{-rt}[XN(d_1) - KN(d_2)] \tag{1}$$

$$P = e^{-rt}[KN(-d_2) - XN(-d_1)] \tag{2}$$

$$d_1 = \frac{\ln(X/K) + (S^2/2)t}{S[t]^{1/2}} \tag{3}$$

$$d_2 = d_1 - S[t]^{1/2} \tag{4}$$

where $N(d)$ = cumulative normal integral

r = risk-free interest rate

S = standard deviation of log percentage change in annualized prices

X = futures price

K = strike price

t = time to expiration, annualized

C = call premium

P = put premium

e = exponent (2.7183)

ln = natural logarithm

The BSM model requires information on five independent variables in order to estimate fair value for a non–income-earning future option:

1. The current futures price
2. Option strike price
3. Days to expiration
4. Risk-free interest rate
5. Standard deviation of futures price change or volatility

The current futures price is an empirical approximation of the mean of the normal curve; it is necessary in order to center the probability distribution. The option strike price represents our interest in a specific futures price outcome along the normal curve. Days to expiration, or the number of trial periods, is the empirical case frequency (or number of trial events) of the distribution. Calendar days are used in the BSM model rather than trading days. The risk-free interest rate is the opportunity cost of capital; it is taken to be the Treasury bill or note for the appropriate term of the option. Finally, volatility is the estimated standard deviation of futures price change over the number of trial periods, expressed as a logarithm. Volatility represents the probability of outcome of futures prices. To convert equation (1) for use with income-earning assets (for example, stocks, bonds, or currencies), one would need to subtract an expected future yield from the expected futures price. Colburn (1990:157–158), and Labuszewski and Sinquefield (1985:117–119), have presented empirical examples of the BSM formula.

Unfortunately, there is some evidence that the assumption of log-normality for financial and commodity asset price change may be inaccurate. Records in the stock and commodity markets over many years appear to show that financial asset price changes have a higher cental tendency with longer tails than the theory of normality would suggest (Cootner, 1964; Brealey, 1969; Turner and Weigel, 1990; Sterge, 1989; Nelson, 1988; Peters, 1991). This tendency for long-term financial assets to deviate from the normal curve is illustrated in Figure 2.4. While there are very few observations in a normal distribution above or below two or three

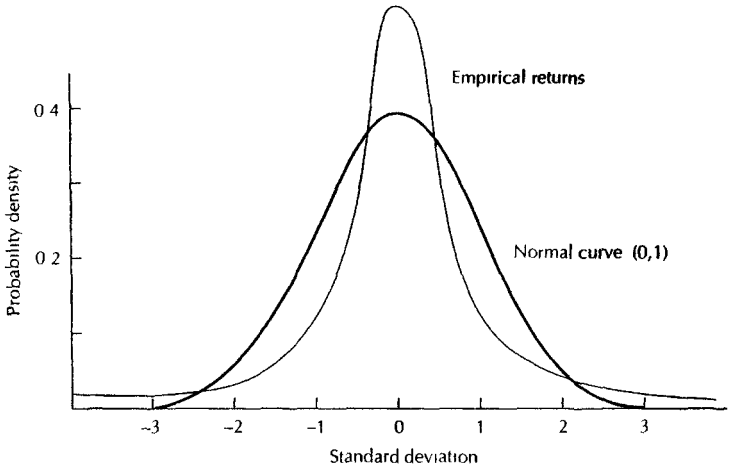

Figure 2.4 Normal distribution and financial asset returns. (Normal distribution from Hastings, 1975.)

standard deviations from the mean, actual long-term financial returns may include observations as much as six standard deviations away, as found by Turner and Weigel and by Peters in stock returns from 1928 to 1988. On the other hand, long-term empirical stock returns show more data points in the center of the distribution than the assumption of normality predicts; these indicate greater *peakedness.*

Nelson (1988) demonstrated the wide variations possible in year-to-year wheat futures price change from 1967 to 1987. These data also show the empirical distribution has a greater central tendency and longer tails than the standard normal curve (Figure 2.5).

Research on stock and asset price change has consistently found that the standard normal distribution does not fit actual events smoothly and Cootner (1964) suggested stock prices may be only approximate standard normal. The normal curve is specifically defined by different measures, or *moments.* The first and second moments are the mean and standard deviation. The third moment of the normal distribution is *skewness* or *tilt;* in these cases the mode is not the same as the mean, and the mode may be either skewed to the left or right. Skewness or tilt is evident in long-term financial asset price change.

The fourth moment of the normal distribution is *kurtosis*, or the degree of thickness in the tails and peakedness of the center; a thick-tailed distribution shows *platykurtosis* while a more peaked

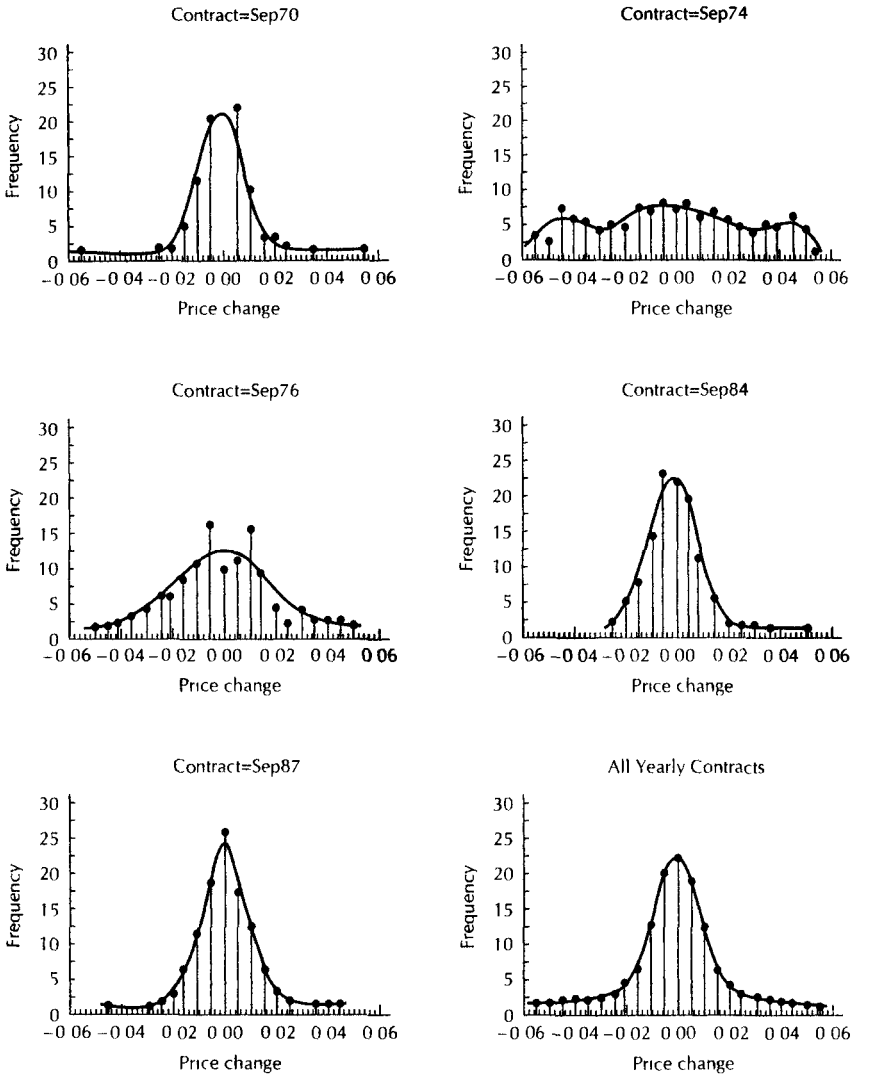

Figure 2.5 *Logarithmic September wheat futures. (Selected and all years, 1967–1987, from Nelson, 1988.)*

distribution shows *leptokurtosis.* Most studies of long-term stock and asset returns show that price changes have both leptokurtic (peakedness) and platykurtic (thick-tailed) features but that neither theoretical curve alone fits the empirical data.

When differences between the actual returns and a standard normal curve are calculated, an error curve is derived. Peters (1991) computed this error curve for actual stock returns from

1928 to 1988 as shown in Figure 2.6. The normal curve was compared with the frequencies of occurrence for Standard & Poor's 500 five-day returns.

Another reason why the assumption of log normality for asset price change has been questioned is that these models cannot easily handle American options, which have the possibility of early exercise. Additionally, log-normal models become increasingly inaccurate as the term of the option lengthens, especially longer than a year (Wong, 1991).

All fair-value models accept the theoretical link between asset price change and probability theory, but some have abandoned the normality assumption and proposed alternative price change models based on different probability assumptions. Alternative explanations of price change seek, in effect, to eliminate the error curve formed as the difference between the normal and the asset price change.

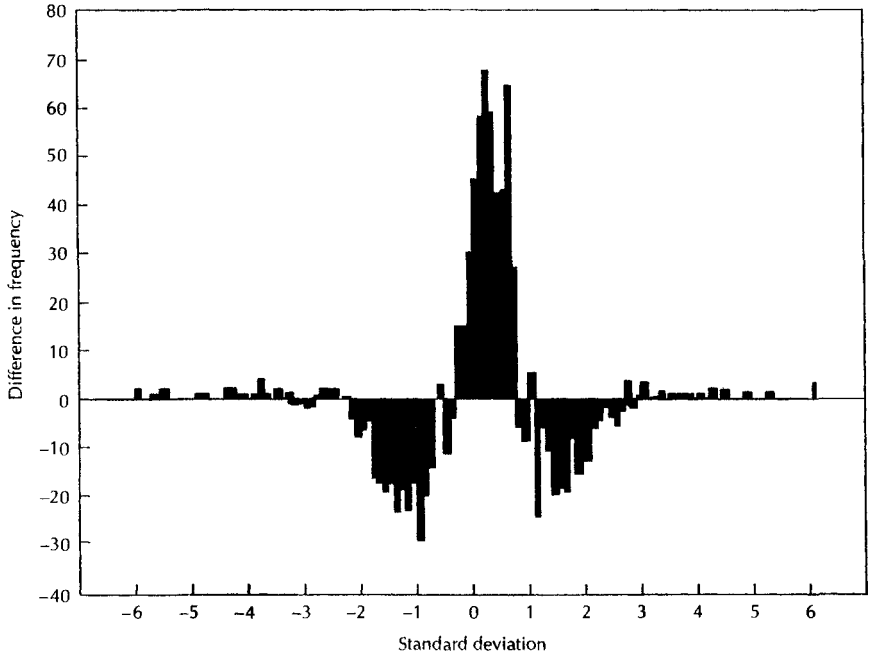

Figure 2.6 Error curve derived as the difference in frequency between the Standard & Poor's 500 five-day returns (1928–1988) and a normal curve. (Copyright ©1991 Edgar Peters. Reprinted by permission of John Wiley & Sons, Inc.)

Mandlebrot (1964) suggested that stock returns may resemble a class of Paretian distributions also known as fractals, which have unstable variance. Peters (1991) shows, however, that a wide range of recent financial asset price changes fit fractal assumptions and models.

The Cauchy distribution (Figure 2.7) has a density function with longer tails than the normal distribution and also tends toward peakedness. Indeed, the Cauchy seems to provide a better fit of long-term wheat futures change than a normal distribution. Unfortunately, the Cauchy has an unstable mean and infinite variance, which complicate its statistical use. Nevertheless, the Cauchy has been used effectively in a number of scientific fields (see Olkin, Gleser, and Derman, 1980).

Recently, fair-value models based on the probability of binomial walks have been proposed, following the work of William Sharpe (1978) who derived the same result as Black and Scholes using only elementary mathematics. These binomial models, also known as lattice-based models, are an exciting area of option research at present. Cox and Rubenstein (1985), who have developed binomial models, also suggest that asset changes may follow some sort of *diffusion-jump* process. Gastineau (1988) has proposed models based on nonnormal distributions of change in option and asset pricing. For European options, binomial models will converge with the BSM model at the limit, but binomial models appear better able to incorporate American option features (Wong, 1991).

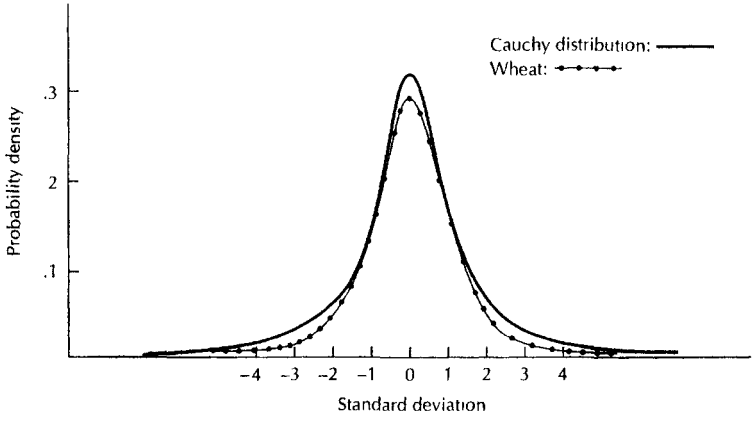

Figure 2.7 Cauchy distribution and wheat price change (1967–1987). (*Source of wheat data:* R. D. Nelson, 1988.)

Both the log-normal and the lattice option models depend upon what one assumes short-term interest rates will be over a long time for back-month options. For this reason, these models are somewhat insensitive to the true *costs of carry* of the underlying asset, which are very important for option prices. An accurate option pricing model, therefore, seeks to take into account both short- and long-term rates of interest, or the term structure. Recently, a number of innovative term-structure binomial models have been put forward for bond options and bond option, traders may wish to study these models more closely (Wong, 1991).

All option traders should be familiar with the statistical assumptions of the model they are using and feel comfortable with them. Most option traders and market makers, however, are traders, not statisticians. To aid the understanding of basic concepts, this book will continue to use the BSM model for illustration, but some modifications of this model will be introduced in later chapters so as to fit prevailing market conditions better.

Fortunately, the BSM model may often be used by option market makers for practical trading to produce reliable results since making option markets is more a question of relative pricing than absolute pricing. Market makers are not so much interested in what real or true fair value may be as in which options are mispriced relative to each other under the same model.

Moreover, one can introduce sophisticated modifications to the BSM model, if needed, by incorporating a strike skew function, which will be discussed elsewhere. For these reasons, the BSM model is used as the statistical basis of this study, and the model will be considered *robust* to the violation of the assumptions of normality. A partial list of software that may be used for calculating option fair values from the BSM model is included in the Appendix.

VOLATILITY

The standard normal fair-value models, such as the BSM model, depend upon five independent variables or unknowns for non–income-earning asset options: interest rates, asset price, strike, expiration date, and standard deviation. Volatility in the BSM model is just the standard deviation of asset prices around the mean over a one-year period. If an underlying asset has a 10% volatility and current price of 100, then there would be a 68% chance (assuming

a normal distribution of price changes) that the asset price at the end of the year would be within the range of 90 to 110.

The general formula for the standard deviation is:

$$SD = \sqrt{\frac{\Sigma(P - \overline{P})^2}{N}}$$

where SD = standard deviation

P = price

$\overline{P}$ = mean price

N = number of trials

Here is an example:

(P)rice	$(P - \overline{P})^2$
90	100
100	
110	100
Mean = 100	$\sum$ = 200

The standard deviation, then, would be $\sqrt{200/3} = 8.16$. In this example, one standard deviation is 8.16 above or below the mean of 100. The probability of an event being within one standard deviation on one side of the mean is about 34 percent, or within one standard deviation above or below the mean is about 68 percent. In this example, the final price over one year will be within 91.84 and 108.16 about 68 percent of the time.

To avoid the possibility of negative asset prices, standard deviation is customarily calculated using natural logs of rates of change in lieu of absolute difference. In this case, the formula for the standard deviation is expressed as a rate of change.

$$SDV = \sqrt{\frac{\Sigma(\ln R - \overline{R})^2}{N - 1}}$$

where $R = \frac{P_{(t+1)}}{P_{(t)}}$

Computation of the SDV gives a percentage change of the standard deviation and is what is meant by volatility. This volatility is always historical, since it depends for its data upon actual past market fluctuations. SDV may also be called historical volatility (HV). Historical volatility may be used to derive the standard deviation in absolute points (SDP) through the general formula:

$$\text{SDP} = P \times \text{HV}$$

For example, if historical volatility is 0.10 and the current price is 100, then SDP will be 10 points over one year. Knowing how to compute the standard deviation in points is useful to know, as will be evident in later chapters.

To adjust for shorter time intervals than one year, volatility must be divided by a time factor. This factor is the square root of the number of periods in a year ($\sqrt{PY}$). For example, the weekly time factor is composed of 52 periods, and $\sqrt{52} = 7.21$. Common time factor adjustments (PY) are:

Daily	$\sqrt{250} = 15.81$
	$\sqrt{365} = 19.10$
Weekly	$\sqrt{52} = 7.21$
Monthly	$\sqrt{12} = 3.46$
Bimonthly	$\sqrt{6} = 2.45$
Quarterly	$\sqrt{4} = 2.00$

The BSM model calls for the use of calendar days (365), not trading days (250) to expiration, as a time adjustment, but market practitioners often use the number of trading days as the adjustment factor. Is it reasonable to assume that all calendar days are equal for calculating price change, or is the number of trading days a better adjustment divisor?

The answer to this question is a straightforward empirical matter. If the futures price change between Friday's close and Monday's close is equal to or greater than the price change within an average consecutive three-day trading period, then calendar days carry implicit trading volatility risk that should be taken into account. If the weekend price change is equal to or less than an intraweek daily price change, then there would appear to be no weekend volatility effect, and thus the time between Friday's close and Monday's open could be treated as a one-day change, justifying

the use of a 250-day yearly period. In either case, the answer is easily settled empirically for any specific market.

In summary the formula to find the standard deviation in points (SDP_t) over period PY is:

$$SDP_t = P\left(\frac{V}{\sqrt{PY}}\right)$$

For example, if V is 15 percent and P is 100, then the standard deviation over one day is 0.95 points, using 250 calendar days.

$$0.95 = 100\left(\frac{.15}{15.81}\right)$$

If one knows the point standard deviation change over a year and wishes to calculate volatility, the above formula is solved in this manner:

$$V = \frac{SDP}{P}$$

Thus, a 1.25-point SD over a year for a commodity priced at 100 indicates that volatility is 0.0125, or 1.25 percent.

To find the volatility (V_t) over any period PY:

$$V_t = \frac{SDP_t}{P}\sqrt{PY}$$

$$V_t(\text{in percent}) = V_t \times 100$$

This relationship is sometimes also known as the $T^{1/2}$ Rule; variance is equal to the standard deviation times the square root of time. For example, the historical volatility over one day is 0.382 (or 38.2%) if SDP daily standard deviation is 2.00 and P is 100:

$$.382 = \frac{2}{100} \times 19.1$$

Given the importance of volatility in option pricing models, alternative measures of expressing volatility have been variously proposed. Instead of using only price change from daily close to close, it is possible to use a volatility measure based on daily high and low prices. Conceptually this measure makes sense since differences in day-to-day price closes may be flat even though large

price volatility occur intraday. If the underlying asset is regularly traded in a liquid market, the high/low estimate of volatility may be more accurate than the close/close volatility estimate for identical time periods (Bookstaber, 1987).

In particular, a price change measure of volatility may not work well with bond options, since bond prices converge to par at maturity. For debt markets, yield change has been found to be a better measure of standard deviation in bond option models (Wong, 1991). Finally, one must remember that these volatility and standard deviation measures will only work as long as actual price change can be described by a normal curve.

Standard deviation, or volatility, is perhaps the most important variable in the BSM model. While four of the five variables in the BSM model all are easily known at any one time (futures price, strike, expiration, and interest rates), volatility is not. Thus, even if the distribution of futures price changes tends to resemble a log-normal curve over time, it is still necessary to know the width of the curve (that is, its standard deviation) before one can calculate an option's fair value. But here a problem arises: What measure of volatility is to be used?

When discussing price volatility one must distinguish among three different aspects of volatility: future, historical, and implied. Future volatility is the standard deviation that would be calculated from the present to some period in the future. In a sense, this is the real volatility. For the BSM formula to work as theoretically intended, the measure of standard deviation that one must use is the future (or real) volatility. Obviously, what is of interest is not what has happened, but what will happen.

However, since the future is unknowable, so is future volatility. Since the fair-value model cannot work without knowledge of future volatility some estimate must be made in order to use the BSM formula. There are two ways of estimating what future volatility may be: use either historical or implied levels of volatility.

Historical volatility is the empirical standard deviation of futures prices from some time in the past up to the present. Since it has already happened, historical volatility is always known with certainty. The only judgment to be exercised concerning historical volatility is the period to be selected. Half-year, 90-day, 20-day, and even 10-day periods have been used. However, each of these different periods is likely to provide a different estimate of volatility level, and indeed there may be no single best estimate of historical volatility. Generally, long-period estimates are the most stable and

shorter periods the most unstable and erratic because the latter are subject to immediate market conditions.

The use of historical volatility in the BSM model (for whatever period chosen) would not present a problem for option fair-value determinations if future volatility levels happened to coincide with past volatility levels. If what has happened will always happen, then future volatility can be estimated directly from historical levels confidently.

Unfortunately, not only is there no a priori reason for past and future volatility to be the same, but also the record shows they usually are not, whatever period is used. Future volatility often changes over time, rendering historical volatility a poor and unreliable estimator of future volatility, whatever the period of historical volatility used.

In statistical terms the difference between past and future volatility is known as *heteroscedasticity,* or the variance of the variance over time. This difference represents a problem to the uncritical use of estimators based on the assumption of statistical normality in any time series model. The existence of severe heteroscedasticity over commodity cycles is intuitively well known; commodity prices are sometimes quiet for extended periods, and then break out in extreme price runs in one direction or another, before returning to quiescence again. Historical volatility is a poor predictor of future volatility.

Implied option volatility may be used as an alternative to historical volatility in estimating future volatility. Whenever an option is traded at market price, it is possible to work backward in the BSM formula and solve for volatility. The resultant *implied* volatility is the market's current estimate of future volatility, and is an alternative to using historical volatility as a forecaster of future volatility.

There is little evidence, however, that current implied levels of volatility are reliable or accurate estimates of future volatility To see this intuitively, suppose that "real" future volatility will be indicated in the 20-day historical volatility lagged backward 20 days. Current 20-day historical volatility, for example, is only the future volatility 20 days ago. When historical volatility is lagged and then compared with the level of implied volatility at that time, there is little indication that implied levels will necessarily equal real volatility. Sometimes implied levels fall below real future volatility and sometimes they are above, as illustrated for October cotton futures and options in Figure 2.8.

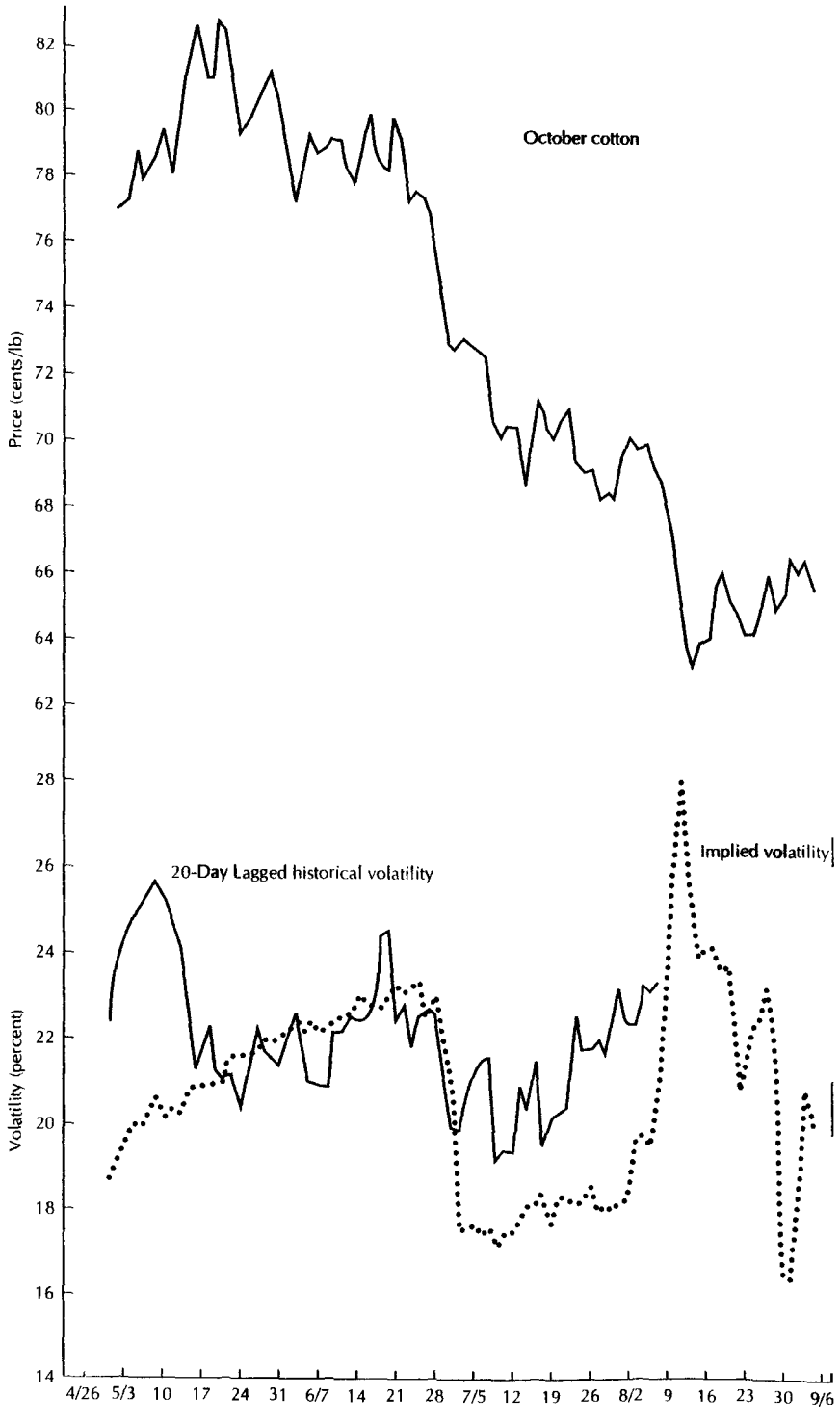

Figure 2.8 Real and implied volatility for October cotton futures and optior prices. (*Source:* Mr. Tom Bertolini, New York Cotton Exchange.)

Generally, there is no consistent short-term pattern between real future volatility and implied levels in most option markets. Since real or future volatility may never be known until after the fact, one must estimate future volatility from either unlagged historical volatility or current implied volatilities in the market in order to derive accurate fair-value option prices. If neither is a reliable empirical predictor of future volatility, then the robustness of the BSM option pricing model is weakened.

For the most part, this uncertainty is not a serious impediment for market makers to using the BSM model. Prudently profitable market making does not depend, for the most part, on knowledge of true future volatility and, therefore, true fair-value prices, but rather on relative pricing. In practice, market makers almost always initially set bid/asked prices around recent implied volatility levels, and only use historical levels for reference.

FINANCIAL AND FUTURES OPTIONS

The discussion of options thus far has been generally restricted to options on non–income-earning spot instruments, or to all options in general. Nevertheless, options will display different characteristics depending upon the differences of the underlying asset: stocks, stock indexes, bonds, currency, commodities, or futures on these. One important distinction among option markets is the cost of carry for the underlying asset, whether positive or negative.

For most financial assets (stocks, bonds, and currency) there is an initial positive cost of carry, with income from dividends, yield, or interest. In the case of stock dividends, for example, where the dividend will reduce the price of the stock by that amount, the current price of the stock must be discounted by this dividend amount when considering the value of the in-the-money or intrinsic option at expiration. For example, if a stock priced at 100 is paying a $1 dividend quarterly, then the 90 day option with strike of 100 will be on a stock worth only $99 after dividend payout, assuming no further price changes by expiration. What is of importance for the option is not necessarily the current stock price but the present estimate of expected future value. In this example this estimate would be $99 and, in effect, the forward price.

For some assets, such as commodities like gold or copper, there is a negative cost of carry due to costs of storage, insurance,

shipment, and so on for physical assets. This negative cost of carry will tend to raise the forward price. In the bond market in particular, where bonds are often financed by other bonds, both positive and negative cost-of-carry positions are possible. For example, if a holding in long-term U.S. Treasury bonds is being financed by a sale of short-term Treasury bills, there will be a positive cost of carry if bill rates are below long rates, but a negative carry if short rates exceed long rates. In summary, differences between income- or yield-producing assets (stocks, bonds, currency) and non–income- or negative-earning assets require that option models be adjusted for asset price cost of carry.

Options on real underlying assets and options on futures also need to be distinguished. For example, an option on stocks or stock indexes is a call or put right that corresponds to an ongoing similar underlying asset. In this sense, a January call is a right on the same stock asset as a May call. However, futures options are rights to futures contracts and not to the underlying asset itself. A January futures call holds a right against a different underlying asset (the January futures contract) than a May call (the May futures contract). Since futures options introduce new complexities into option market making, let us briefly review futures markets here.

A futures contract is the right to take delivery of a specified quantity and quality of a commodity (spot) at the monthly expiration specified in the contract. Thus, if a trader is long (bought) a December sugar futures, at a specific day in September, the trader can take delivery of 112,000 pounds of a specified grade of sugar at a certified warehouse. If a trader is short (sold) a similar futures contract, then he or she must stand ready to deliver spot at expiration. A futures contract of itself does not specify any price at which delivery is to take place. Rather, the purchase (or sale) cost of the contract, when first purchased or sold on the market, becomes the basis cost to buyer or seller. In modern futures markets, contract expiration and delivery dates are sequenced in cycles every quarter or several months apart.

In economic theory, futures markets are useful because they allow for efficient price discovery and for investment risk shifting, or price insurance. Futures markets provide an efficient and competitive price-setting mechanism ("discovery") by allowing prices

to respond immediately to shifting supply-and-demand conditions through an ongoing open outcry auction market. Futures markets also provide a risk-shifting function to producers, merchants, and industrial consumers, allowing industry to shift future price-change risk to other commercial interests, dealers, or speculators. This risk shifting allows business to achieve neutralization of risk and thereby improve trade efficiency.

For example, a farmer will typically sell forward to a merchant all or part of his or her crop that has not yet been harvested. This sale protects, or hedges, the price of the crop while it is still being grown. This action helps avoid the price pressures associated with seasonal harvesting and temporary oversupply. Mining or petroleum producers also use futures contracts in this way, to hedge against oversupplies.

Likewise, a merchant will sell forward if shipping to the destination market requires a long time, exposing the commodity to unexpected price shifts before sale. In the early days of forward trading, a merchant would buy cotton in Savannah and then ship it north to New York, before reshipping it to New England or Liverpool textile mills. By selling forward, the merchant protected his investment in inventory during the long lag between purchase and final delivery. Hedging crude oil during its long transport to market is a more recent example.

Also, manufacturers who use commodity raw products will typically hedge their cost of inventory during production by either selling or buying forward. Cotton mills today, much like the nineteenth century, still have routine recourse to forward contracts in controlling inventory costs.

Futures markets in the nineteenth century and on into the twentieth century were dominated by trading in agricultural products and precious or industrial metals; these remain a significant proportion of all modern futures trading. In the last several decades, however, currency, bond, and stock index futures have been traded, and they now account for the largest proportion of all futures trading. Worldwide, the growth of futures trading has been explosive as the financial industry learned that futures trading may be used to manage and hedge portfolios of bonds, stocks, or currencies.

Futures need not and usually do not trade at the same price as the underlying spot commodity; the difference in price is referred

to as the *basis*. If the underlying commodity is a non–income-earning asset that must be stored somewhere, the futures price is usually higher than the spot commodity price in order to account for the cost of storage or negative carry. For non-income stored negative carry markets, futures prices will trade over spot (positive basis), and the more distant futures will trade over near futures prices, to account for the requisite costs of storage and insurance. When the distant futures price exceeds the near futures price, the market is said to be in *contango*.

If the near futures price is higher than the distant price, however, the market is said to be backward (Figure 2.9). *Backwardation* in the futures time spread for non–income-earning durable commodities may come about for several reasons. In agricultural commodities there is often a normal backwardation between crop year futures cycles, which comes about because the storage costs of the new crop are unrelated to storage costs of the old or current crop. If the new crop has not yet been harvested, no storage costs have been incurred; and the far distant futures contract may trade, therefore, lower than the less distant or near contracts. Also, in both seasonal and nonseasonal commodity markets, backwardation may occur as a result of severe supply shortages or sharp increases in unfilled demand. For example, although gold and silver futures normally trade in contango, during the 1980–81 attempted silver corner by Bunker Hunt, precious metals futures went backward.

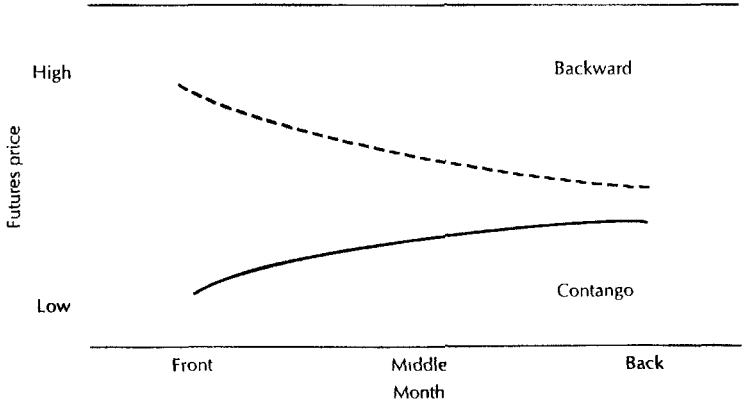

Figure 2.9 Futures contango and backwardation.

For income-earning assets—stocks, bonds, or currency—the futures time spread may be somewhat different from that of non–income-earning assets. For financial assets, the cost of carry is not negative, and may even be positive. Stocks and bonds earn dividends or interest, and do not entail large costs of storage. For assets with a positive cost of carry the futures time spread is normally backward. In effect, the future price is discounted by the positive yield on the asset over time, and thus distant futures prices may trade lower than near-term prices, all else equal. Consider, for example, the situation in which one-year Treasury notes, par 100, yield 5 percent over one year. If one year forward futures contracts are priced at 95, they would approximately discount the positive cost of carry and trade at normal backwardation. Positive carry futures markets, however, also may display contango time spreads at times, for different reasons. Further discussion on time and basis spreads of futures may be found in works listed in the bibliography (Williams, 1986).

Futures options have rights over a futures contract, not the underlying asset. This distinction introduces important new risks to option traders, who make markets in futures options time spreads. The risk does not exist for stock option traders, however. It is also possible to have option markets on both the underlying asset and the futures on the underlying asset. For example, there are options markets on stock indexes at the same time that options on the futures of the stock index are being traded in a different market. This situation opens up the possibility of intermarket option arbitrage. This study will emphasize futures options as the general case, with discussion of differences arising with stock or bond options where appropriate.

3

Option Risks

INTRODUCTION TO OPTION RISKS

Fair-value option models link statistical theory to investment risk, which modern capital theory defines as the probability that the actual return from an investment will differ from the expected return (Hagin, 1979; Gehm, 1983). Defined this way, investment risk is useful in evaluating portfolios that have equal profit but differ in risk taken. The chances of doubling one's money on a single toss of the dice in a casino promises a better rate of return than the interest earned in a bank account, but runs radically different risks. Capital may seek the largest profit, but at the "best" risk/reward ratio.

The link between probability theory and investment risk makes it possible to quantify option investment risk in very precise ways. Except for strike, any change in the other variables in the fair-value option model (interest rate, futures price change, days to expiration, and volatility) may bring about changes in option prices over the duration of the option cycle until expiration. Thus, these variables represent *risks* to an option portfolio.

This chapter briefly characterizes each fundamental risk to an option's price. These risks are defined as delta (Δ, futures price change), gamma (Γ, change in delta), theta (Θ, time decay change), kappa (K)/vega (implied volatility change), and rho (P, interest rate change). Vertical spreads across single calendar strikes also are subject to a *skew risk* in the implied volatility strike spread. Holding calendar or time spreads exposes option positions to

additional risks: time delta (spread futures price change) and time kappa/vega (spread option implied volatility change). In addition to these market-based risks, there are also extramarket financial risks.

For the purpose of illustration risk to an option will generally be quoted in dollar-and-cent amounts for a single hypothetical option in this text. However, the real dollar risk must be multiplied by the value of the actual option contract, which will vary depending upon the type of option. For example, a 10-cent gamma risk may not seem large when one is considering only one option, but this 10 cents must be multiplied by the dollar value of the actual option on the asset or futures contract. For example, if the futures option or asset moves $500 for every whole futures point, a 10-cent risk is really a $50 risk for one option ($0.10 \times $500), and a 10-option position would have a $500 gamma risk. Likewise, a single stock option represents an option on 100 shares of stock and a 10-cent risk on a stock option is really a risk of $10.00.

In trading options it is easy to lose your shirt if you do not understand the variety of risks involved. These risks will be illustrated in the following sections using simple option positions.

DELTA (Δ) RISK

The delta risk of an option is a ratio reflecting the dollar amount of change in an option price for every dollar change in the underlying futures price. The delta risk is also known as the *hedge ratio.*

$$\Delta = \frac{\text{Dollar change in option price}}{\text{Positive dollar change in asset price}}$$

When an option's value increases with a positive change in asset price, delta is positive. If an option's delta is +.30, for example, the option will increase in value 30 cents for every positive dollar change in asset price. If an option's delta is −.30, then it will decrease in value 30 cents for every positive dollar change in asset price.

Figure 3.1 shows the payoff of a long 100 strike call and its delta over a range of futures prices from 90 to 110 at 60 days, 30 days, and one day to (or at) expiration (with implied volatility levels at 15 and interest at 10 percent).

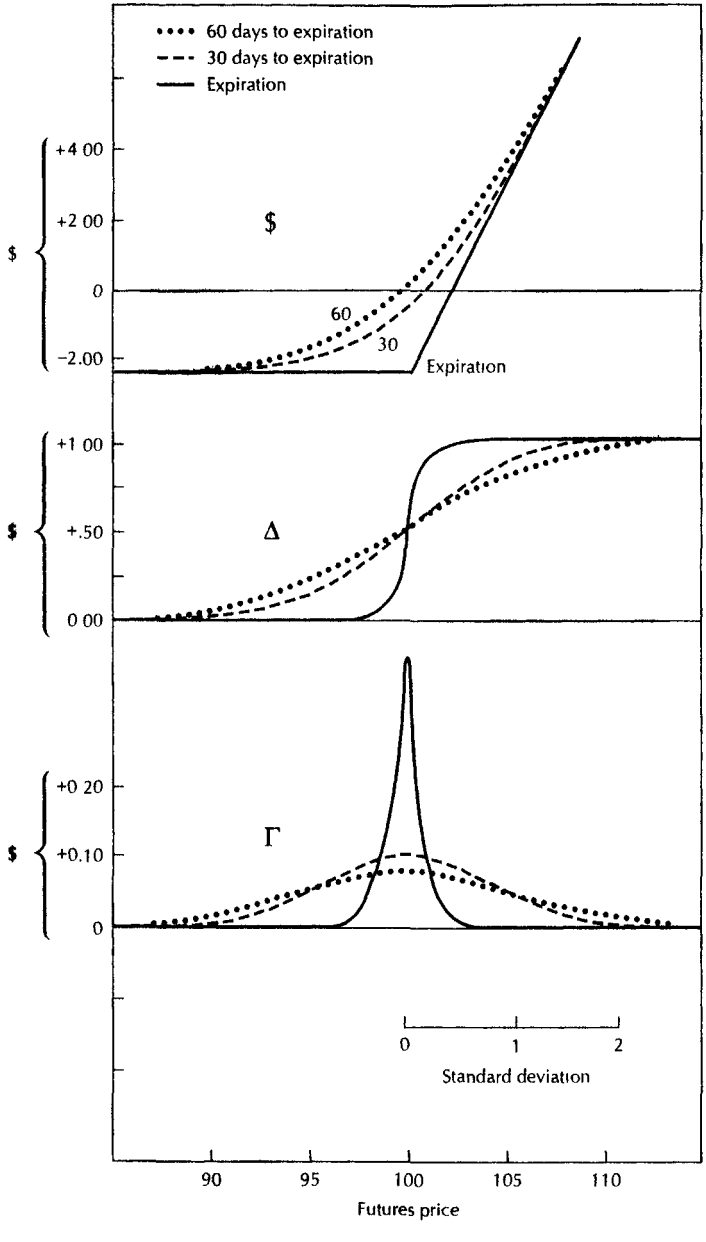

Figure 3.1 Delta and gamma risks of a long 100 strike call and its payoff.

When futures or the asset itself are at the strike of the call, the delta of the long call always is a +.50. It can range upward to +1.00 when futures are above the strike and to zero when futures are below the strike. A short call would have a negative delta, since it would be the inverse or opposite side of the long call. The long put has a negative delta and the short put a positive delta. The delta of these and other option positions are discussed in more detail in Chapter 4.

A very deep in-the-money option will usually have a delta close to 1.00, that is, the option price will move almost exactly as the asset price itself. This correspondence makes intuitive sense, since a deep in-the-money option is likely to expire with the value of the underlying asset and, therefore, will move in value almost equally. Likewise, a very far out-of-the-money option usually has a very small delta, which is asymptotically approaching zero. This pattern also makes intuitive sense, since a far out-of-the-money option is likely to be worthless at expiration and to be relatively unresponsive to underlying asset price changes.

An at-the-money option will always move in price about half that of an asset contract, for a delta of $\pm$.50, reflecting the equal probability of an option expiring in or out of the money. Generally, no single option can have a delta greater than +1.00 or less than −1.00 since no single option can change, dollar for dollar, more than a single asset contract at expiration. The underlying asset itself will always have a delta of +1.00 or −1.00, depending upon whether a long or short position is held.

The passage of time increases or decreases the delta of an option. An in-the-money option will show an increasing delta and an out-of-the-money option a decreasing delta. Option delta, therefore, *drifts* in time. This delta drift becomes more pronounced as the option expiration approaches reflecting the time shrinkage of the standard deviation, all else being equal.

In statistical terms, an option's delta is also the probability that at expiration the futures will settle above the option strike, making the option in-the-money. A delta risk is a futures or asset price-based risk. A high positive delta indicates that in-the-money options are likely to remain in-the-money, and a low positive delta indicates the likelihood that the option will expire worthless. At-the-money options with a delta of +.50, therefore, have a 50:50 chance of expiring in- or out-of-the-money.

A positive delta indicates a bullish option position and a negative delta reflects a bearish one. A long call or short put is a positive delta position, and a short call or long put is a negative delta, for example. The delta of a total option position is simply the net sum of the delta's of all separate positions.

Some option traders and market makers, however, often prefer to remain neutral about market direction, that is, *delta neutral.* To create a delta neutral option position, a trader must offset the deltas of the separate options. For example, if the delta of an at-the-money option is always +.50, then an option position that is long $2 at-the-money calls, and short $1 future, will be delta neutral, that is, $2 \times .50 - 1.00 = 0$. A long or short straddle can also be a delta neutral position if the strike of the straddle is at the current futures price, for example, $(+.50) + (-.50) = 0$. The Δ's of many single-month option strategies will be found in Chapter 4.

GAMMA (Γ) AND LAMBDA (Λ) RISK

An option's delta is not a constant, but changes as the asset price changes and makes the option more or less in-, at-, or out-of-the-money. The change in an option's delta as the asset price moves up or down is referred to as gamma risk.

$$\Gamma = \frac{\text{Net change in } \Delta}{\text{Dollar change in asset price}}$$

While delta is a measure of risk to price from asset price change, gamma is a measure of the risk due to the stability of the delta. As the asset price increases $1.00, the delta of an out-of-the-money long call will increase, for example, from .30 to .35. This .05-cent increase in the delta is the gamma of the call position at that futures price range, and represents the added risk to an option when delta changes.

Gamma can be negative as well as positive. A long call or long put always has a positive gamma, and a short call or short put always has a negative gamma throughout the entire futures price range, with a single modal peak centering around at-the-money options. The deep in- or out-of-the-money wing options have the least change in delta and, therefore, the lowest gamma (Figure 3.1).

For example, at 60 days to expiration, the long 100 call still has some positive probability of expiring in-the-money and, therefore, will have a positive delta and a positive gamma over the range of futures prices from 90 to 110. When futures are at strike (100), gamma is 0.075 cents at 60 days to expiration; gamma rises to 0.10 cents at 30 days, but with a smaller range.

By one day to expiration, however, the same option that was once very deep out-of-the money has virtually no chance of expiring in-the-money, and its delta and gamma have shrunk to zero. By one day to expiration, the gamma of the long 100 call will have exploded to over 0.25 cents when futures are at strike, but very deep in-the-money options will have a high positive delta but gamma at zero since delta has reached its upward limit of one. Generally, at-the-money options show an increase in gamma as expiration approaches, while the deep in-the-money or out-of-the-money option gamma decreases over time. The gamma risk of many single-month option positions is found in Chapter 4.

Another measure closely related to delta is the *elasticity,* or lambda, of an option position; lambda is defined as the percent change in the value of an option position relative to the percent change in the value of the underlying asset.

$$\Lambda = \frac{\text{Percent change in option value}}{\text{Percent change in asset price}}$$

The lambda of an option will always be greater than one, since an option is a leveraged instrument relative to the underlying asset. A deep in-the-money option with little time remaining to expiration will have a lambda approaching one, reflecting equal proportional changes between option and asset. As an option becomes out-of-the-money, its lambda will become much greater than one, indicating its higher leveraged power for gain or loss relative to the underlying security.

THETA (Θ) RISK

The change in option price due to the days remaining to expiration is known as time decay, or theta risk.

$$\Theta = \frac{\text{Change in option's value}}{\text{One-day change in time remaining to expiration}}$$

This risk should be obvious since, all else being equal, an option contract with fewer days remaining is worth less than an equivalent one with more days to expiration, for the extra days add value. Thus, option prices tend to decline as expiration approaches, and indeed, decline more rapidly the closer the expiration is.

For example, the theta risk of a short 100 call at 60 days to expiration is only +2.5 cents per day when futures remain at the strike price, which means that the trader will profit by this amount for one day by time decay. At 30 days, the short call will have a theta just under +4 cents, which will rise to over +10 cents on the last day before expiration so long as futures remain at a strike of 100. If there is a move in either direction, however, the in- or out-of-the-money option will have a smaller theta risk (Figure 3.2).

If an option trader is long an option, each day brings some decrease in the option's value, all else being equal, and the trader is exposed to negative theta. Short options will have a positive theta, for the short option seller will profit from the time decay value of an option. To be *short premium* is to have a positive theta, because the trader expects to make a profit from time value decay. Generally, at-the-money options with the least amount of time remaining to expiration will experience the largest positive or negative theta values. The time value decay is slowest with long trading life remaining in the option.

Note that the theta of a long call is strikingly similar in form to its gamma risk over this range of asset prices, except that theta is about half the magnitude of gamma and of opposite sign. The identity of risk between gamma and theta in single-month options will be important in the evaluation of risk in the more complex option positions and strategies that are discussed in later chapters.

KAPPA (K)/VEGA RISK

Even if there is no change in futures or asset price risk (delta or gamma) or in time risk (theta), an option price may be affected by changes in the market's valuation of implied volatility. This valuation is formally referred to as kappa, although popular usage also refers to this risk as *vega*. Kappa /vega risk is one of the most important risks to an option price and is defined in this manner:

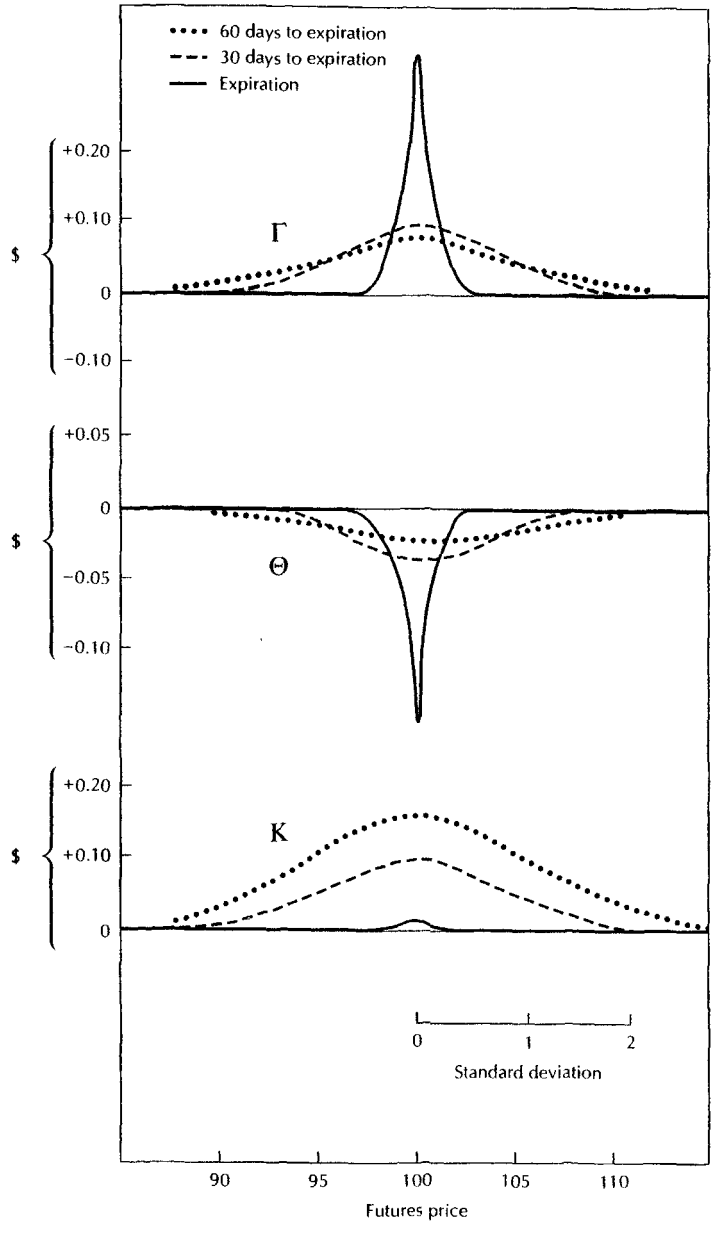

Figure 3.2 Gamma, theta, and kappa/vega risks of a long 100 call.

$$\text{K/vega} = \frac{\text{Dollar change in option price}}{\text{Positive one-point implied volatility change}}$$

For example, an increase in the implied volatility from 15 to 16 (one point) would increase the value of a long 100 strike call by a positive 16 cents at 60 days to expiration when futures are trading at the strike price (Figure 3.2). The kappa/vega of the long call would be positive over all futures price ranges but would move either upward or downward as the call became in or out of the money. Of course, there is no limit on the increase in implied volatility. Were a 10-point increase in implied levels to happen, the long (short) call would have a profit (loss) of $1.60.

This amount must be multiplied by the actual value of the option contract to determine the precise dollar risk. For a single stock option on 100 shares of stock, the amounts above must be multiplied by 100; and for a futures option that has a value of $500 per contract point, the above kappa/vega amounts would be multiplied by that full point value. Thus, a $1.60 kappa/vega risk in the above example would represent an $800 risk to the holder of one futures option and $8000 to the holder of 10!

An option position will have a positive, negative, or neutral kappa/vega risk at any single asset or futures price. Positive kappa/vega risk will result in increased profits from increased implied volatility, and negative kappa/vega will result in a loss from increased implied volatility change. A long call is positive kappa/vega and a short call, negative. A long put is also positive kappa/vega and a short put, negative. To be *long volatility* means that a trader has a positive kappa/vega option position.

Remember, kappa/vega risk does not vary directly with delta risk; they must be treated separately. Implied volatilities of options may move independent of futures price changes in either price direction. For purposes of illustration, let us suppose an important announcement (such as weekly USDA export sales) is scheduled for the following day; the price directional implications of this announcement are unknown. Option prices might rise on an increased market estimate of the possible future price change after the announcement, and might fall just as rapidly after the announcement if futures price change did not occur, all else being equal. The increase and decrease in option prices before and after this announcement reflect changes in the market's implied volatility price.

Paradoxically, sudden sharp futures price changes (up or down) may occur in conjunction with falling implied volatilities. This divergence might seem counterintuitive, since if real volatility is up, one might suppose that implied volatility must be up also. However, one would be ignoring that implied volatility is the market's estimate of future volatility, not historical volatility. An increase in current volatility accompanied by a fall in implied volatility merely suggests that the market expects future volatility to fall, however high it currently is.

It is not uncommon for option traders to be long a call and find, when futures prices do move sharply higher, that their call has increased only marginally in price even with a moderate positive delta. While such traders are likely to be upset and perhaps even blame the market maker for somehow cheating them out of their profit, what the option traders do not realize is that the price paid for the call had already been based on a high-volatility estimate. Once the move had occurred, the market believed no further moves were likely and lowered its estimate of volatility. Although the option traders were correct about price movement, they were whipsawed by the market's changing estimate of volatility. These traders did not understand kappa /vega risk (see letters to *Barron's*, April 18, and July 18, 1988). One exaggerated example of the possibility for a trader to be correct about the direction of the market but still lose money in options because of an implied volatility change occurred in the stock market crash of October 1987, when short out-of-the-money call options rose in price after the 500-point drop in the market, reflecting a market-implied volatility level of well over 100!

In practice, the implied volatilities of options are one of the major risks to any option trader. Much analysis, evaluation, and strategy are devoted to kappa/vega risk.

RHO (P) RISK

One of the variables in the BSM futures option-pricing model is the risk-free rate of interest. This interest rate represents the cost of carry of an option position, or the opportunity cost to capital of trading in options, that is, what unoccupied capital may safely earn. A positive cost of carry earns interest, and a negative cost of carry incurs interest payments. The appropriate risk-free rate

of interest is taken from the yield curve of the U.S. Treasury bill market that matches the option cycle. If interest rates change, the cost of carry and the value of an option will also change, all else being equal. Change in the cost of carry that leads to change in the value of an option is referred to as rho risk.

Consider a futures call which has a hypothetical fair value of $100 and a zero cost of carry, that is, the cost of capital is free. The purchase or sale of the fair-value call at $100 results in neither profit nor loss over the long run. But if capital is not free, the buyer of the $100 call must deposit cash to purchase the option (a negative carry). Thus, he or she forfeits the earned interest on the $100, and the seller gains the use of these funds, which may be deposited to earn interest after having satisfied margin requirements. If the call is to have a fair value that includes the cost of carry, the call will sell at some discount to $100 that equals the cost of interest rate carry.

Therefore, as interest rates increase, the fair value of futures options falls, all else being equal. For example, the fair value of a 100 call with 200 days to expiration, a volatility of 15, and a risk-free rate of interest of 8 percent, is $4.24. If interest rates were to rise from 8 to 15 percent, however, the value of the option would drop to $4.08. The 16-cent difference represents the discount on $4.24 of an additional 8 percent annual interest prorated for 200 days.

For nonfutures options, such as stock options, an increase in interest may either increase or decrease the value of the option. An increase in interest will increase the carrying cost of long stock positions, making a call option more desirable than holding stock. Likewise, an increase in interest will make shorting the stock more attractive than holding a long put, in which case the value of the put will decline.

The impact on option prices of a change in the risk-free rate of interest is not so large as that of some other option price risks. Interest rates rarely move so dramatically over six-month periods as the previous example might imply, although the early 1980s did witness extraordinarily high historical interest rate changes. In some foreign option markets these interest rate changes may also not be unusual.

Interest rates, nevertheless, remain an important market risk and necessitate sophisticated strategies for market makers. These will be discussed in Chapter 5.

SKEW RISK

The central implied volatility of an option strike series is that found in the level of the at-the-money straddle. But if the 100 call is trading at an implied level of 15, what about the 105 call, or any other options of this cycle, whatever the strike?

All strikes of one calendar series need not trade at the at-the-money implied level. A 100 call may be trading at a 15 volatility, but a 105 call at a 20 implied volatility. This difference in the implied volatility levels of single-cycle options across strikes is termed the *implied volatility slope,* or *skew.*

Positive skew refers to both out-of-the-money puts and out-of-the-money calls (wings) that are trading at higher volatility levels than at-the-money options (center). For a flat skew, all strikes trade at approximately the same volatility levels. In a negative skew the option wings trade at lower volatilities than the center strike options. Since puts and calls each have different out-of-the-money/in-the-money volatility spreads, there is a put skew and a call skew in every single-cycle strike series. These hypothetical skew relations are shown in Figure 3.3.

Skew risk exists in any spread option position because the degree of implied skew may change. If a skew goes from positive to flat to negative, vertical spread traders are affected, as changes in skew affect option spread prices. If an implied volatility skew exists, then it generally is a linear or curvilinear function. Sometimes saw-toothed and choppy skews exist temporarily as a result of transient imbalances or trades on close. These are relatively unimportant and may safely be disregarded.

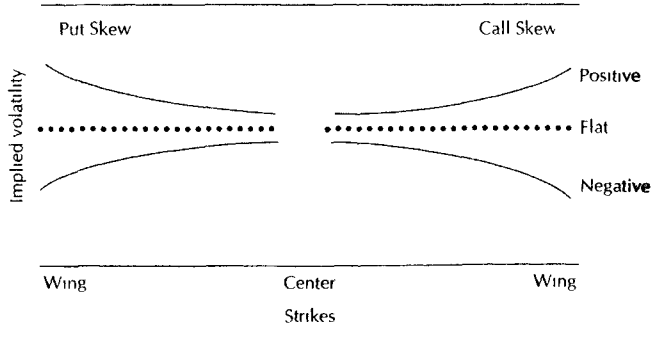

Figure 3.3 Implied volatility skew.

On many if not most futures and financial option markets some form of positive put and call skew exists. In practice, option traders tend to value out-of-the-money options higher than the BSM model suggests. In other words, option market participants trade as if a more frequent or extreme price volatility than log-normal probability suggests is likely. From Chapter 2 we know that actual long-term returns of financial assets have greater extremes than would be expected by the normal distribution, and that these are consistent with implied volatility skew.

Positive skew patterns are also consistent with assumptions of popular option trading strategies. Option speculators may prefer to be long out-of-the-money options, owing to their relatively low absolute cost and low absolute risk of loss as well as their infrequent potential for big payoffs (for example, the lottery ticket play). Conversely, option traders may prefer selling at-the-money options because of the greater absolute time decay (positive theta) profits. These market demand situations may tend to lower the implied volatilities of the center and raise those of the wings.

The only common exceptions to the general pervasiveness of positive skews in futures and financial option markets are the sometimes negative call skew in stock index futures options, and the sometimes negative or flat skew in commodity futures puts. Stock option participants devalue out-of-the-money calls and some commodity option markets devalue out-of-the-money puts with negative skews. The topic of skew risk will be taken up again in Chapter 8.

TIME SPREAD RISK

The discussion of option risks so far has been limited to single-month positions only, either as single options or as some type of vertical spread. *Time spreading* means holding a simultaneous position in more than one calendar month; it is also known as a *horizontal* or *calendar spread.* The option leg that is the closest to expiration is termed the *front month,* and the option leg furthest from expiration is known as the *back month.* An option leg between the front and back months would be the *middle month.* Time spreading neutralizes some of the risks of holding single-month positions. At the same time, calendar spreads are subject

to other time-based risks that may be as severe or more severe than those risks neutralized. The two time-based risks for calendar spreads are termed *time* delta and time kappa/vega *risks.*

Time delta risk is defined as the delta risk to any futures option time spread caused by a change in the underlying futures time basis. Time delta risk is only applicable to futures option positions, and not to cash or stock option positions, since the underlying instrument never expires in stock calendar option spreads (that is, the shares of common stock), while futures price time spreads reflect differential prices of monthly contracts. Time delta risk is obvious to any futures option trader, of course, but less obvious sometimes to traders who only trade stock or cash option spreads. Stocks and the cash market do not have a time basis. Futures and futures options do, and this is the source of time delta risk.

All option calendar spreads, whether futures or nonfutures, are subject to time kappa/vega risk as well. A position composed of two different contract months risks having the implied volatility spread of the different months' change. The implied volatility of a time spread is by no means constant, and exposes a time option spread to price risk, sometimes more than the risk neutralized in a single-month strategy.

Both time delta time kappa/vega are extremely important and will be discussed further in Chapter 6.

EXTRAMARKET RISKS

The risks to the value of an option discussed so far are generally known and may be quantitatively measured. There is another class of option risks that are less well known or measurable. They are the extramarket risks that frequently result from financial irregularities or calamities of one sort or another.

A financial risk to all options and futures traders is the possibility that a trader may lose his or her entire capital in a clearinghouse suspension or bankruptcy. Market makers and exchange option traders are required to keep trading funds in an exchange-regulated clearinghouse that guarantees the trades of its members. Although the clearinghouse system protects a trader from the default of any individual trader with whom he or she trades, it does not protect an exchange trader from the default of the clearinghouse itself. Any floor trader takes the risk that another

trader at his or her clearinghouse may suffer losses so large that the clearinghouse itself becomes temporarily or permanently insolvent. This risk is real, and over the years there have been several stock or futures clearinghouse failures, such as Volume Investors in 1985, Fossett in 1989, and Stotler in 1990. Others are bound to happen in the future. Fortunately, in these clearinghouse suspensions virtually no floor traders lost account capital, but of course, it is not impossible that this might happen in the future.

An attendant risk in a suspension of a trader's clearinghouse is that the trader will not have the means to trade the carryover position, exposing the trader to market risk. If a trader is inadvertently long or short in the market and unable to adjust his or her position because a clearinghouse suspension has frozen all accounts, the trader may incur steep losses whether or not the clearinghouse eventually remains insolvent.

Some means do exist to lessen extramarket risks. For instance, the general public may be somewhat safer trading through large brokerages; and stock and currency options are guaranteed settlement through the Options Clearing Corporation. Nevertheless, the large over-the-counter market in options, especially bonds and currencies, has no clearinghouse guarantee system. Therefore, traders are exposed to what is known as *counterparty* risk, which exists when the opposite party to a contract defaults without clearinghouse guarantees. In this case, hypothetical profits turn into losses.

Extramarket risks should be considered carefully by prudent option traders, however improbable. Although they are rare and freak, these risks may be devastating to a market maker's or dealer's capital.

4

Position Risk Profiles

BASIC OPTION POSITIONS

To evaluate option position risks comprehensively, one first must have some working classifications of the universe of possible types of option positions. The basic option positions may be considered to be composed of the 22 core positions listed in Table 4.1.

These positions may be defined briefly as follows (see the Appendix to this chapter for some illustrated examples):

Straddle. Long (short) same strike put and call; for example, long the 100 call and the 100 put.

Strangle. Long (short) different strike put and call; for example, long the 110 call and the 90 put.

Vertical spread. Long (short) and short (long) different strike options; for example long the 100 call, short the 110 call. Vertical spreads may be either bull or bear spreads depending on the directional expectation of price.

Fence. Long (short) put/call and short (long) a different strike call/put; for example, long the 100 call, short the 90 put.

Ratio spread. Long (short) one strike option, and short (long) a greater number of other strike options; for example, long one 100 call, short two 110 calls. There are four possible ratio spreads: long call, short call, long put, and short put. A ratio spread is long or short in the direction of the net options held. For instance, a long call ratio spread is short one 100 call, long two 110 calls. Long ratio spreads are also known as *backspreads.*

Table 4.1 Basic option positions

1	Long call
2	Short call
3	Long put
4	Short put
5	Long straddle/strangle
6	Short straddle/strangle
7	Bull spread
8	Bear spread
9	Bull fence
10	Bear fence
11	Ratio spread long calls (call backspread)
12	Ratio spread long puts (put backspread)
13	Ratio spread short calls
14	Ratio spread short puts
15	Long cartwheel
16	Short cartwheel
17	Long wrangle
18	Short wrangle
19	Long butterfly/condor
20	Short butterfly/condor
21	Conversion/reversal/box
22	Time spreads

Note: See Chapter Appendix for payoff and risk profiles of single-month positions.

Cartwheel. Long one ratio spread and short another ratio spread; for example, short one 100 call, long two 110 calls, long one 100 put, short two 90 puts.

Wrangle. Long (short) both the put and call ratio spreads; for instance, short one 100 call, long two 110 calls, and short one 100 put and long two 90 puts.

Butterfly. Long (short) the middle strike options and short (long) an equal number of outer strike options on both sides of the middle; for example, long one 90 call, short two 100 calls, and long one 110 call. A *condor* means selling (buying) the middle strikes as a strangle rather than as a combination of a straddle. A butterfly or condor is always both a *bull* and *bear spread* combined.

Synthetics. A conversion, reversal or box. Synthetics are generally risk-free option positions composed of some combination of underlying instrument and same-strike puts and calls. These terms are defined and discussed in Chapter 5.

Time Spreads. A time spread is an option position composed of two or more different option calendar cycles (the "legs"). For further details see Chapter 6.

For a simpler classification, this text groups straddles and strangles as one position type, and butterflies and condors as another single type. Straddles/strangles and butterflies/condors are grouped together because members of each pair are essentially similar to each other except for the distance of the strike spread. This study considers the standard vertical ratio spreads as separate positions, and examines several position types less frequently discussed: fences, cartwheels, and wrangles. Conversions, reversals, and boxes may be treated as one position type from the standpoint of risk and are, therefore, one synthetic type.

All time spreads are considered as belonging to one category in Table 4.1. Actually, any time spread may be created in any of the other basic position types, as either a two-legged or three-legged (butterfly) time spread. When time spreads are considered, the number of basic possible positions increases by a factor of 21 for each leg added, since this is the number of the basic single-month positions possible. Because of the complex possibilities of time spreads, this topic will be taken up separately in Chapter 6. This chapter will only discuss single-month option position risks.

There are other option positions, not listed in Table 4.1, that are irregular in one way or another. Irregular option positions are usually some combination of one of the basic option positions, executed separately over a very wide strike/futures range. For example, the payoff schedule of an irregular option position might look like Figure 4.1.

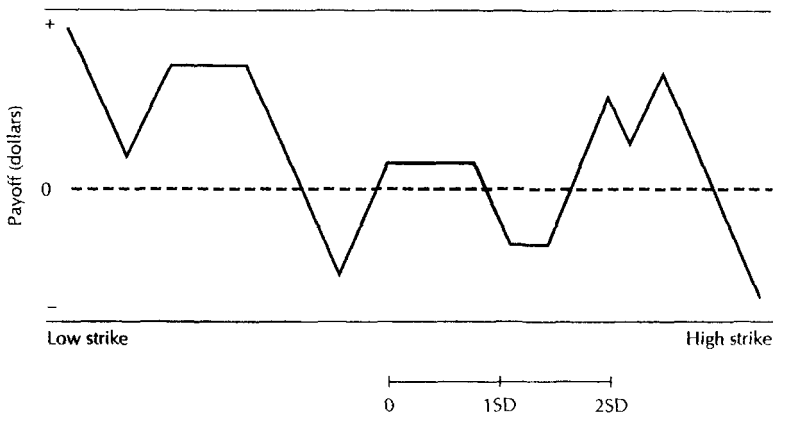

Figure 4.1 Irregular option position expiration payoff.

Irregular option positions are *quirky* and display unstable risk profiles over extremely wide strike or futures price ranges. Essentially any irregular position may be built up with some combination of the 20 basic single-month positions spread across wide strike ranges. *Quirkiness* results from a tendency for a spread to take on the risk characteristics of the nearest strike option to futures near expiration as the absolute standard deviation shrinks as expiration nears. For example, a butterfly position in the back month may take on the risk characteristics of a straddle in the front month as the standard deviation shrinks over time. Almost any extremely large option position that is composed of almost all the strikes traded will have some quirky characteristics very near to expiration. This topic will be reserved until expiration risks are discussed in Chapters 5 and 7.

Dealer positions are rarely composed of only single options or simple spreads. In practice, a market maker's total carryover position over the course of an option cycle is likely to grow quite large and may easily total thousands of contracts in complex spreads, even if few contracts are traded every day. There are several reasons for this inevitable growth.

First, a trader making markets in active months will be faced with a market that potentially has up to 50 or more option instruments differing by strike, month, and option type. One reason a large *inventory* of options accumulates over the course of an options cycle is simply that if a market maker can make a market in all option series, he or she will probably do so with enough

capital to maximize profits. Over time, an active market maker's total carryover position will probably cover every strike that has an open interest. Second, an option carryover position is likely to grow larger with time because the market maker usually makes necessary continual small adjustments to correct for risk changes and *drift*.

But no matter how large or complex any option carryover position becomes, it will always resemble (on a single-month basis) one of the basic option positions in Table 4.1, irregular option positions not withstanding. An option trader should always know the exact type of the carryover option position, no matter how large or complex.

POSITION RISK PROFILES

Options differ from underlying assets in their exposure to risk. Financial assets or futures generally are only subject to positive or negative price change risk. Options also are subject to this risk as delta risk; in addition single-month options are exposed to gamma, theta, and kappa/vega risks. Options that include calendar spreads are exposed to all of these and to time delta and kappa/vega risk as well. The payoff at expiration and at 30 days and the delta, gamma, theta, and kappa/vega risk at 30 days are shown for selected single-month option positions in the Appendix to this chapter.

In Chapter 3 option risk was characterized by sign and dollar size for any small increment in instrument price, gamma, implied volatility, or time. But the sign and size of this risk may, and usually do, change as assets or futures price change. For example, a long call position may have a large negative theta in dollar terms at-the-money but a small negative theta either deep in- or out-of-the-money. A vertical spread may have positive theta at the short option strike but negative theta at the long option strike.

That option risks change as asset or futures prices change is termed the *risk modality* of the option. Each option position has its own risk modality for delta, gamma, theta, and kappa/vega. To quantify risk modality, one must measure risk over the extended range of futures prices over which options may continue to change value. This range must be at least six standard deviations wide to be able to capture extreme change.

The change in sign of risk in the basic single-month positions is reflected in the number of peaks (statistical modes) of gamma, kappa/vega, and theta risk (see Figure 4.2). No change in sign corresponds to one peak or mode, characteristic of single-option positions or straddles. A change in risk sign will be reflected in a twin-peaked risk (one positive and one negative) and is bimodal. A second change in risk sign results in a triple-peaked distribution and is trimodal.

These risk sign modes show that any spread risk is not equally proportional to the distribution of risk over a range of futures prices. Risk tends to gravitate to that strike option in a spread nearest to the futures price. The different modal peaks, then, just reflect the number of underlying spreads with different risk characteristics within a total position.

A single nonspread option or straddle position has unimodal kappa/vega, gamma, and theta risk profiles. Bi- or trimodality is introduced to risk any time any vertical spreads other than straddles are traded. Consider a simple bull spread in the figure on page 76, where gamma, kappa/vega, and theta risks have become bimodal. On the upside, a bull spread is positive theta and negative gamma and kappa/vega. On the downside, it is negative theta and positive gamma and kappa/vega. The two modes are positive or negative at the downside and upside strikes of the spread. Bimodality of gamma, kappa/vega and theta risk is also characteristic of other spreads, including fences, ratio spreads, and cartwheel positions.

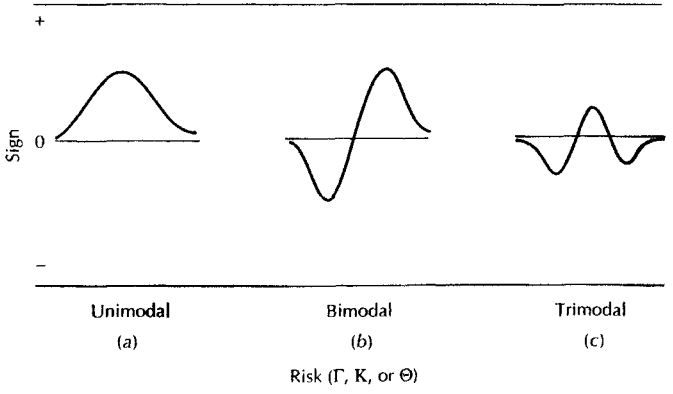

Figure 4.2 Risk sign modalities.

and an implied volatility for the 100 call at constant 15 with the 100 call valued at 2.40. At an implied volatility of 15 (flat skew) the 105 call has a fair-value of 0.74; at an implied volatility of 16 (low positive skew) the 105 call has a fair-value of 0.86, and at an implied volatility of 17 (high positive skew) the 105 call is worth 0.98.

The fair-values of the 100–105 (1×1 and 1×2) call spreads under these different skews are found in Table 8.7. If a trader were to buy the 100–105 (1×1) call spread (buy the 100, sell the 105 calls) when skew was flat at 1.66, then a loss could accrue if the skew went from flat to low or high positive skew ($1.54 − $1.66 = −0.12 and $1.42 − $1.66 = −0.24). If the trader were to short the ratio spread by buying the the 100 call and selling the 105 call twice (1×2), the results financially would be even worse as skew moves from flat to low to high positive (0.68 − 0.92 = −0.24 and 0.44 − 0.92 = −0.48).

From the standpoint of skew risk, it is not a good idea to be a buyer of bullish call or bearish put spreads when skew is flat, or to attempt to short the ratio spread. Rather, selling option (1×1) spreads and going long the ratio spreads during flat skew periods is a better strategy. During periods of high positive skew relatives, it is better to buy option spreads and cover any short spreads or long ratio spreads. A long wrangle will experience a profit going from a flat to a high skew but a loss going from high to flat skew.

For large traders (with many hundreds or thousands of contracts and spreads), a skew wobble in the implied relatives can translate into thousands of dollars of fluctuations in an option account balance on a daily basis. Spread positions will experience

Table 8.7 100–105 Call spread values by skew

	Spread (dollar value)	
Skew	*(1×1)*	*(1×2)*
Flat	1.66	.92
Low positive	1.54	.68
High positive	1.42	.44

(See Table 5.3 for underlying option prices.)

wobble fluctuations in daily profit and loss accounts. Failure to take into account wobble can mean a market maker taking on spread trades that will later prove overpriced even when there is no change in the at-the-money implied volatility or futures prices!

The best way for a market maker to ensure that this does not happen is to be aware of the ranges for skew relatives on the particular option contract in which he or she is making a market and attempt to spread trade accordingly. A trader would try not to pay more for wing options than flat or low skew prices, and would attempt to sell or liquidate positions during high skew prices, all else being equal.

A further difficulty in skew risk is that the center strikes of the skew will change in a large asset or futures price move, thus moving or shifting the entire skew itself, as in Figure 8.4.

A shift in skew may significantly affect the value of vertical spreads and is a risk to the complex spread trader. For example, if a trader executed a short 100 strike and a long 110 strike call spread when skew is centered at 100 strike then a futures/asset price move to 110 will shift the entire skew so that the short call is now at the high point on skew and the 110 call will be at the center strike. In effect, even though skew remained positive after the shift, the call spread would have suffered the same financial effects as if skew had gone from positive to negative!

The effects of skew wobble occurring simultaneously with skew shift may be additive and present some significant risk to any complex option vertical spread trader. Moreover, there is no perfect way to hedge complex vertical spreads against skew wobble and shift risk, short of synthetic trading. Nevertheless, although it is

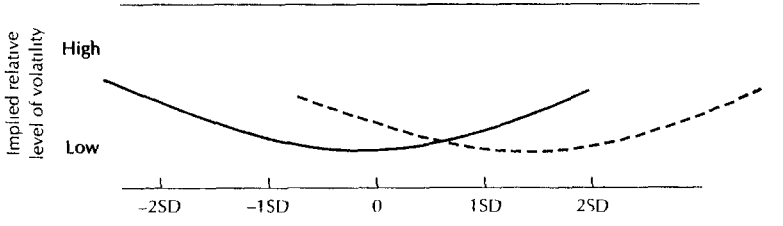

Figure 8.4 Skew shift resulting from a large asset or futures price move upward.

an important risk, skew risk is not unlimited, fortunately. In the end, market judgment and cautious experience may be the best guides.

TRACKING TRADING RESULTS

A market maker will usually be able to tell whether he or she is trading successfully by whether he or she is making a profit over the long run. Clearinghouses will provide a trader with a daily net account balance that gives the final tally. It is useful, however, for a market maker to keep his or her own financial accounts, independent of clearinghouse records. There are several reasons for doing this.

First, keeping independent accounts can be used to verify clearinghouse records. Although intentional mistakes are rare, clearinghouse records often show clerical errors, sometimes of a large magnitude. Without an independent account record, a trader is sometimes helpless to correct clearinghouse statements. If the clearinghouse statement remains uncorrected, which sometimes happens, a trader may incur a large windfall loss.

Second, without an independent account to highlight clearinghouse account inaccuracies, a trader will not know his or her exact position and cannot adjust properly or know the risks he or she is under. There will always be traders on any exchange who do not know their exact position because of clearinghouse confusion, and who then take a large hit, sometimes even being wiped out.

Third, keeping an independent account gives the trader the ability to track and analyze trading results. Although one might assume that overall profitability is an indication of market-making success, this need hardly be the case. Market-maker profitability must be distinguished from overall profitability if a market maker wishes to assess accurately his or her own trading success (or loss).

For accounting purposes, total profit/loss (P/L) may be divided into that P/L resulting from the carryover position, and that P/L from daily trades. To show the daily P/L for the carryover position, mark the value of the position to settlement prices at end of Day 1 and Day 2 without adding new trades, and take the difference. This will represent the P/L due to the carryover position over one day. To show the daily P/L due to new trading, subtract the prices

of the option at trade from the settlement prices for that day. The daily P/L due to the carryover position when added to the daily trade P/L will equal the total P/L of the position.

The net daily carryover P/L corresponds to the P/L of the limited-risk strategy itself, while the daily trade P/L will show the results due to market making. A market-maker will be trading successfully if the market-making profits are a large or the largest share of total profits over the long run.

Consider the P/L separately for the daily carryover account and the daily trading account, as in Table 8.8. The total profit for the combined carryover and day trades in Table 8.8 is +1, but this is entirely contributed by the day trade results. To do the next

Table 8.8 Example of total profit/loss statement for a hypothetical market maker

Daily Carryover Account				
Carryover and Prices	*Settlement Day 1*	*Settlement Day 2*	*End Day 2 Profit / Loss*	
100 short 100 calls	2.00	2.10		−10.
100 long 100 puts	1.00	.90		−10.
Long 100 futures	.80	1.00	+20.	
Subtotal profit			+20.	−20.
Total carryover profit			zero	
Daily Trading Account				
Day Trades and Prices	*Day 2 Trade Price*	*Settlement Day 2*	*End Day 2 Profit / Loss*	
Sell 10, 100 calls	2.20	2.10	+1	
Buy 10 futures	100	100	zero	
Subtotal profit			+1	
Day trading profit			+1	
Net Profits by Activity				
Carryover position			zero	
Day trades			+1	
Total net profit			+1	

day, the net day trades by strike and option are added to or subtracted from the carryover position, and a new carryover position is established for the next day carryover position accounting, plus whatever next day trading occurs.

A trader may wish to refine the analysis of P/L for the carryover position further by doing each cycle separately. With an ongoing daily record of this accounting, a more detailed P/L record is visible. For example, consider the daily P/L accounts in Table 8.9. In this Table, the net result of the May, July, and December positions is −$100 ($0 − 150 + 50), largely stemming from the poor July cycle results. This result is not necessarily too serious since many monthly position P/Ls will show short-term deviations that balance out over time. A trader would, of course, want to examine the larger July loss in this example to see whether it represents a poorly hedged position or just reflects daily irregularities in settlements, which quickly adjust themselves.

In Table 8.9, the net result of the carryover positions was −$100, but profits from daily scalping were $300, giving a total net profit of $200. With daily market making contributing a large share of gross and net profits, one may infer that this trader is market making successfully but might be able to improve results by adjusting the carryover position.

The accounting assumption that day trade net results are attributable to market making over the long run is probably valid. Although the daily net trading profits may also be attributable to gamma adjustment tactics, over the long run these daily gamma adjustments should net to zero against daily carryover position changes, leaving the remainder of the daily P/L as net market-making profits. Thus, it remains true that a positive net P/L total over the long run on the daily trading account is an approximate

Table 8.9 Daily P/L by component

	Carryovers ($)				
Day	*May*	*July*	*December*	*Day Trading Net ($)*	*Net Total ($)*
1	−100	−200	+300	+ 50	+ 50
2	+100	+100	−200	+250	+250
3	0	− 50	− 50	0	−100
Total	0	−150	+ 50	+300	+200

indication of market-making profitability if a trader has remained delta neutral in the carryover position.

If over a longer period a trader's total profits are largely contributed by day trade results, then a market maker is trading well. If this is not the case, or the trader is losing money on net balance, then looking at the accounting results separately by day trade and carryover position will help diagnose the problem by indicating where loss is occurring.

COMMON MISTAKES

Mistakes are inevitable. Since market makers and scalpers work for a small arbitrage profit, however, even small mistakes can mean the difference between profit and loss. Although mistakes are inevitable, they should be minimized as much as possible.

What are the most frequent mistakes option traders make? There are at least four areas in which traders probably err repeatedly. These are:

1. Failure to hedge delta
2. Incorrect delta hedging
3. Overtrading
4. Poor kappa/vega risk adjustment

Failure to Hedge Delta (Δ)

Not maintaining delta neutrality is probably the single most frequent mistake an option scalper will make. The best type of option scalping is spread trading, but when this is not immediately available an option trader must hedge delta to remain delta neutral. Many traders, however, do not immediately execute delta adjustments in the futures market after doing a nonspread option trade. This is a mistake. *Taking a shot* is, in fact, speculation, and market makers who do not hedge their delta risk immediately are speculating.

The lure to speculate by not futures hedging one's option trades immediately is strong. By attempting to get the edge on the futures

side, a market maker can often double the profit of the trade. This will happen just often enough to yield positive reinforcement for the trader to continue to avoid immediately hedging futures.

In the instances in which the trader does not get the edge, and does not get his or her futures hedge off, the market may (and probably will) move against the trader on the unhedged option side, causing losses instead of profits. If an option trader were willing to accept a loss immediately, the losses and profits would probably balance out to neutral in the long run, perhaps giving the trader the false impression that speculation is, at worst, neutral.

But what often happens is that an option trader is not willing to take a small loss to hedge when it will mean a small loss on the original trade. Eventually these small losses turn into large losses. In speculating by not hedging immediately or at all, many option traders cut their profits short, but let their losses run. For these traders, market-making profits are eaten up very quickly in speculative losses.

The best course is to avoid speculation, and to hedge delta whenever any option trade is completed that is not an option spread trade. If a trader waits to hedge delta in the best way, the outcome may be large losses. *Hedge when you should, not when it is best.*

Incorrect Delta (Δ) Hedging

Poor delta hedging entails trading options at prices that are unrealistic compared with the current futures price at which delta hedges may be traded. In other words, hedges cannot be done at posted futures prices because the board prices for some reason are lagging behind actual pit trading. If a market maker is limited to using futures prices as posted on the board to price options, he or she will be at a disadvantage and risk trading options at unrealistic futures prices. There are several ways to avoid this trap.

One way is to stand physically close enough to the futures pit to hear futures trading. If an option trader is able to do this, then he or she can hear, and even do, futures trades while trading options. But in large futures and options rings, where there is greater physical distance between the futures and options pits, it will not always be possible for an option trader to be near the futures ring. In this situation, a clerk may be necessary to signal

by hand futures prices to the trader and to relay futures orders to the futures pit. This must be considered an added cost of business.

Another aspect of unrealistic futures pricing should be mentioned. It is best not to be last in a string of large option trades in the pit, in order to avoid being in an infavorable futures hedging position. Brokers will sometimes come into the ring and immediately do a large quantity of single strike options spread among different traders. However, these trades are not done exactly simultaneously. Traders who trade the last of a large lot order will find themselves at a competitive disadvantage when attempting to complete delta futures hedges because the impact on futures prices will already have been caused by the delta hedging of the first of the serial traders. What was a large profit for the first traders to do a string trade turns into a large loss for the last of the string traders, after futures prices have been run up or down by the first traders doing their futures hedges. Trading last on a string should be avoided whenever possible.

Overtrading

Another common mistake of option traders is overtrading, or impatience. Some market makers may get so bored from too much inactivity or not participating in trades that trades are done at net fair value. The market maker is not waiting for a sufficient edge to trade. Overtrading is unnecessary, and it may be wasteful if losses accrue because the profit margin is too thin.

A market maker should not look to trade all trades being done in the pit, but only to trade those at his prices. Of all trades done in the pit, there will be some minority of trades done at excessive discounts or premiums to fair value. Good option trading is waiting for these trades during the day, and not overtrading. There is no virtue in being the most active trader, only the most profitable. A good trader must be patient.

Poor Kappa (K)/Vega Risk Adjustment

A trader may become too long or short kappa/vega in context of the market liquidity on the opposite side. A trader does one side of an option trade repeatedly without a corresponding option

spread trade. Even though the trader adjusts for delta neutrality, kappa/vega risk becomes lopsided. In the worst situation a trader has allowed a negative kappa/vega position to develop. This should always be corrected immediately.

The rule of thumb is to remain net long kappa/vega but not to get net kappa/vega more than may be neutralized without too large a loss if necessary. This amount is sometimes difficult to determine in practice and may be a judgment call in which errors will inevitably occur. To a considerable extent, judgment will be based on the trader's appraisal of the net flow of pit/broker option trading.

Other Mistakes

One should also appreciate that clerical errors (mistakes in quantity, strike, month, or side) do occur, and they are truly inevitable. Perhaps one in 20 times, a small error is made. These are a cost of doing business and should be looked at philosophically.

More generally, traders will be making a mistake if they do not learn to relax during the trading day and take vacations periodically. One of the advantages of trading limited-risk carryover positions is that a trader never needs to worry about financial catastrophe happening to him or her and may sleep restfully at night.

Periodic vacations are also in order, and may be conveniently arranged to begin after some expiration. Even if a trader does not liquidate his or her entire position, it is possible with planning to reduce position gamma to very low levels so that one need not be in daily contact with the market.

9

Observations from the Floor

There [on the exchange floor], 50 or 60 grown men stood leaning in each others' faces shouting as loudly as they could in primordial tones of voice that I had never heard before, except maybe the one time I tried to take a shrimp away from my cat, Paco. Veins bulged in the men's necks. Their eyes seemed ready to leap from their sockets at any moment. They waved their arms threateningly at one another in some incomprehensible code.

It was like a cockfight without the cocks. A crowd this angry couldn't help but come to blows at any moment. . . . My heart was in my throat. What was going on here?

The men were trading cotton.

Albert Pacelli, 1991

A first-time visitor to any major commodity exchange will see a horde of screaming, elbowing traders packed densely into an open space as big as a sports stadium, while a deafening roar painfully presses the eardrums. What is a homely workplace to some would count as Dante's Seventh Circle of Hell to most. Recently, several books reported for the first time on the strange rituals of exchange floor social and professional life (Tamarkin, 1985; Kleinfield, 1983; Marks, 1988).

Becoming a good option scalper requires several skills and assets. Good options trading is preeminently a numbers game and requires a good numbers sense. An ability to do quick calculations repetitively, sometimes in one's head, is a distinct advantage. Option traders frequently enjoy playing number games (bridge, chess, go, poker) and are good at them, although few seem interested in

gambling outside of work aside from occasional small but popular informal sports pools.

Physical stamina and the reflexes of an athlete may be just as important as good mathematical intelligence, for floor traders work in a harsh physical and personal environment where competition is intense. Self-esteem is also necessary, for new as well as old traders are likely to suffer direct personal insults on a regular basis or discouragements from other traders who would like to minimize competition. Traders are not known for keeping their voices down or using polite language when they are verbally abusing another trader, fines notwithstanding. There is always someone in any crowd who has difficulty mastering maturity, although the worst offender is likely to be different for each person. Trading pit manners are often nasty, brutish and loud. A new trader must have sufficient self-esteem and perseverance to withstand this kind of occasional psychological warfare and not let himself become emotionally involved.

Correspondingly, a strong cooperative style is a plus, because trading involves being on good terms daily with many traders, brokers, and support staff. Many option traders are partners in option partnerships and must work cooperatively with other partners in a high-stress environment. Traders with bad tempers or those who are unwilling to compromise frequently do not have good floor reputations and risk losing business in the long run.

A sound sense of business conduct and morals must also be considered an important virtue. Most exchanges now require new members to take an ethics course, conducted by lawyers and surveillance officials, that stresses the serious consequences awaiting a trader who breaks the rules. Although the overwhelming majority of floor traders are scrupulously honest, floor traders may eventually encounter some individuals or situations where violating exchange rules may result in some unethical or illegal profit. Unethical and fraudulent traders, however, will almost always be discovered over a lifetime and suffer large fines or expulsion, as they should in every profession.

The professional path of market makers is likely to be varied, and there are several ways to start on this career. For *upstairs* market making in electronic screen-traded over-the-counter financial options, market makers are employees or partners of large investment banks, international banks, or other trading firms, and often start as clerks or assistant traders. Market makers on option ex-

changes are sometimes regulated specialists, as in stock options, or may be independent traders trading for a partnership or their own account.

Capitalization of many of the independent market makers is relatively small. Current full memberships on futures and option exchanges in the United States range from $1,000 to $500,000 (with the New York Futures Exchange probably the lowest, and the Chicago Mercantile Exchange the highest). Other memberships, including special option memberships, may be almost anywhere between those two extremes, although perhaps half of all option memberships in the United States are under $50,000. Sometimes option memberships may be leased, with leases currently running from 1 to 2 percent of option seat value per month. In addition to seat capitalization, option market making requires liquid capital to meet basic margin requirements. This is hardly ever less than $25–50,000 as a minimum. For over-the-counter option markets, capital requirements may be significantly higher to handle large institutional trading.

The 1980s was the second decade of continued strong growth of option trading in the United States. This has been the spur to the expansion of option markets. European and Asian option markets are following the growth path of the United States, promising that the 1990s will be the third consecutive decade of growth in world option markets. Many of these new markets may prove to be among the best option markets for market makers.

John Maynard Keynes once called trading a game. If this is true, it may not be whether one wins or loses that counts, but whether one ever ventures to play. For those who do, good luck.

Appendix

Option Software for Market Making

To keep track of the risks and profitability of large option positions it is almost always necessary to use an option analysis computer software program. Although hand calculations are not impossible, a computer can evaluate instantaneously a constantly changing option position over a number of risk assumptions. Also, software is indespensable for printing fair-value sheets upon which market-maker quotations are based.

Listed below are some option software vendors who appeal to option dealers and market makers. Listing in this Appendix does not imply endorsement of services but is provided for informational purposes only.

TRACK DATA
The Option Group (TOG)
61 Broadway Suite 2301
New York, NY 10006
212-943-4555

COMTEK
141 W. Jackson Blvd. Suite 1531-A
Chicago, IL 60604

EPSILON OPTIONS
Box 3207
Church St. Station
New York, NY 10102

OPTIONS AND ALTERNATIVES SOFTWARE
Chicago Mercantile Exchange Marketing Department
30 S. Wacker Dr.
Chicago, IL 60606
312-930-1000

REUTER SCHWARZATRON
61 Broadway
New York, NY 10006
212-493-7100

SYSTEMS DEVELOPMENT CORP.
RAMBO
141 W. Jackson Blvd. Suite 1240-A
Chicago, IL 60604
312-408-1111

OPTIONOMICS
3191 S. Valley St. Suite 155
Salt Lake City, UT 84109
1-800-255-3374

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